

The parity-separation constant of level- Q divisor weights tends to zero

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Abstract

For an integer N put $Q = \lfloor \sqrt{N} \rfloor$, let q_1 be the least prime exceeding Q , and let $\mathcal{B}$ be the set of squarefree Q -friable integers in $(\lfloor N/q_1 \rfloor, N]$. Let $\kappa(N)$ be the largest average negativity, over the odd- ω part of $\mathcal{B}$, of a divisor weight of level Q that is nonnegative on the even- ω part, normalised by the ℓ^2 norm of its coefficients and by the ℓ^2 norm of the parity defect of the band; equivalently $\kappa(N) = \text{dist}_2(c_{\mathcal{O}}, K_Q) / \|c_{\mathcal{E}} - c_{\mathcal{O}}\|_2$, where $c_{\mathcal{E}}$ and $c_{\mathcal{O}}$ are the divisor-incidence count vectors of the two parities and K_Q is the cone generated by the divisor-incidence columns of the even- ω part. Weighting all even- ω columns equally gives $\kappa(N) \leq 1$, so $\kappa(N)$ measures how much a nonnegative reweighting improves on that baseline. We prove that

$$\kappa(N) \ll_A (\log Q)^{-A} \quad \text{for every fixed } A > 0;$$

in particular $\kappa(N) \rightarrow 0$. The quantitative form of this, with a parameter $\beta \geq 4$ fixed before N grows, is a two-term bound whose leading coefficients are evaluated and whose threshold of validity is not: at $\beta = 6$ it reads $\kappa(N) \leq (2.01317 + \varepsilon) / (\log Q \sqrt{\log \log Q})$ for all sufficiently large N , the constant being recorded as an example rather than as an optimal one. The proof is a direct positive construction: explicit nonnegative weights on the even- ω columns, grouped by the small-prime part $a_Y(\ell) = \prod_{p|\ell, p \leq Y} p$ and truncated at $a_Y(\ell) \leq \sqrt{Q}$. The one arithmetic input it needs is that, after fixing the small-prime part, the two ω -parities of the rough cofactor occur in bounded ratio; we prove this. The elementary observation that drives the signed estimate is that $N < q_1^2$, so an integer $n \leq N$ has at most one prime factor exceeding Q ; the classical large-prime removal is then exact with no upper cutoff on the cofactor, and the required signed two-cutoff estimate follows from a quantitative prime number theorem together with the matching Möbius bound. The result is effective in principle, but the constant implied by $\ll_A$ is not evaluated. The bound is an upper bound only: it says nothing about whether $\kappa(N)$ is positive, nor whether the rate is optimal.

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1 Introduction

1.1 The object

Throughout, N is an integer and

$$Q = \lfloor \sqrt{N} \rfloor, \quad L = \log Q, \quad q_1 = \min\{p \text{ prime} : p > Q\}, \quad \text{thr} = \lfloor N/q_1 \rfloor, \quad (1)$$

with $\log$ the natural logarithm. For $n > 1$ let $P^-(n)$, $P^+(n)$ be the least and greatest prime factors of n , with $P^-(1) = \infty$ and $P^+(1) = 1$; $\omega(n)$ counts distinct prime factors, and μ is the Möbius function. The *band* is

$$\mathcal{B} = \{n \in (\text{thr}, N] : \mu^2(n) = 1, P^+(n) \leq Q\}, \quad (2)$$

split by the parity of ω into

$$\mathcal{E} = \{n \in \mathcal{B} : \omega(n) \text{ even}\}, \quad \mathcal{O} = \{n \in \mathcal{B} : \omega(n) \text{ odd}\}, \quad R = |\mathcal{O}|. \quad (3)$$

Index coordinates by the *rows*, the squarefree $d \leq Q$. With $c_{\mathcal{E}}(d) = \#\{n \in \mathcal{E} : d \mid n\}$ and $c_{\mathcal{O}}(d) = \#\{n \in \mathcal{O} : d \mid n\}$, and assuming $R > 0$, set

$$t_d = \frac{c_{\mathcal{O}}(d)}{R}, \quad p_d = \frac{c_{\mathcal{E}}(d) - c_{\mathcal{O}}(d)}{R} = \frac{1}{R} \sum_{n \in \mathcal{B}, d \mid n} \mu(n), \quad (4)$$

the second equality because $\mu(n) = (-1)^{\omega(n)}$ on squarefree n . For $\ell \in \mathcal{E}$ let $A_{\ell} = (\mathbf{1}_{d \mid \ell})_d$ be its divisor-incidence column, let A be the matrix with these columns, put

$$K_Q = \text{cone}\{A_{\ell} : \ell \in \mathcal{E}\}, \quad (5)$$

and define

$$\kappa(N) = \frac{\text{dist}_2(t, K_Q)}{\|p\|_2}, \quad (6)$$

both norms being Euclidean on the row space. The vector p is the *parity defect* of the band: p_1 is the normalised Möbius sum over $\mathcal{B}$, and p_d its analogue along the row d .

The baseline, and an R -free form. Both $t = c_{\mathcal{O}}/R$ and $p = (c_{\mathcal{E}} - c_{\mathcal{O}})/R$ carry the same factor $1/R$, where $c_{\mathcal{E}} = (c_{\mathcal{E}}(d))_d$ and $c_{\mathcal{O}} = (c_{\mathcal{O}}(d))_d$, and $\text{dist}_2(\cdot, K_Q)$ is homogeneous because K_Q is a cone. So R cancels in (6):

$$\kappa(N) = \frac{\text{dist}_2(c_{\mathcal{O}}, K_Q)}{\|c_{\mathcal{E}} - c_{\mathcal{O}}\|_2}. \quad (7)$$

This form needs $R > 0$ and, as in (6), a nonzero denominator: $c_{\mathcal{E}} \neq c_{\mathcal{O}}$, which is the condition $p \neq 0$ cleared of the factor $1/R$. What (7) removes is the normalisation by R , not either of those two conditions. The quantitative bound $N/R = O(1)$ of Proposition 8 is a separate input, used in the reduction of §4 and nowhere in the definition. Taking the uniform weight $x_{\ell}^{(0)} = 1/R$ on every even- ω column gives $(Ax^{(0)})_d = c_{\mathcal{E}}(d)/R = t_d + p_d$, so $\text{dist}_2(t, K_Q) \leq \|p\|_2$ and

$$0 \leq \kappa(N) \leq 1. \quad (8)$$

This fixes the scale and gives the quantity its reading: *the denominator of (7) is the error made by weighting all even- ω columns equally, the numerator is what the best nonnegative reweighting achieves, and $\kappa(N)$ is the ratio.* What this paper proves is that the optimum beats the uniform baseline by more than any fixed power of log Q .

Two remarks on the definition. First, (4) and (6) require $R > 0$ and $p \neq 0$; Proposition 8 shows that both hold once N is large, and every statement below is for such N . Second, although the level Q enters the name of the constant, (2)–(6) are determined by N and not by Q : the $2Q + 1$ integers N with $Q^2 \leq N < (Q + 1)^2$ share the level Q but need not share the band, although they may (at $N = 10$ and $N = 11$ alike, $Q = 3$, $\text{thr} = 2$ and the band is $\{3, 6\}$). For instance at $Q = 5$ the odd part of the band is $\{5\}$ at $N = 25$, whose row vector is not in the cone generated by the columns of 6, 10, 15, whereas at $N = 30$ it is $\{5, 30\}$ and $t = \frac{1}{2}(A_{10} + A_{15}) \in K_Q$, so that $\kappa(25) > 0 = \kappa(30)$. We therefore write $\kappa(N)$.

1.2 What κ measures

Let $\lambda = (\lambda_d)_{d \leq Q}$, $\mu^2(d)=1$ be a weight of level Q , let $\theta_{\lambda}(n) = \sum_{d \mid n, d \leq Q} \lambda_d$ be its divisor sum, and let $\mathbb{E}_{\mathcal{O}}$ denote the mean over $\mathcal{O}$.

Proposition 1. *Let N be such that $R > 0$. Then*

$$\text{dist}_2(t, K_Q) = \max\{-\mathbb{E}_{\mathcal{O}}\theta_\lambda : \|\lambda\|_2 \leq 1, \theta_\lambda(\ell) \geq 0 \text{ for all } \ell \in \mathcal{E}\}.$$

If $t \in K_Q$ the maximum is 0, attained at $\lambda = 0$. If $t \notin K_Q$ it is positive and attained at a unique λ , which satisfies $\|\lambda\|_2 = 1$.

Proposition 1 is proved in §3. It says that $\kappa(N)\|p\|_2$ is how negative, on average over the odd- ω part of the band, a level- Q divisor weight can be while remaining nonnegative on the even- ω part, per unit ℓ^2 norm of its coefficients. We call κ the parity-separation constant of level- Q divisor weights in the ℓ^2 normalisation. The constraint is one-sided in the parity: nonnegativity is imposed on the even- ω part only. The normalisation is ℓ^2 in the coefficients λ_d , not Selberg's quadratic form and not an ℓ^1 norm; the feasible cone is the same for every norm, the constant is not. And the level is fixed, so for each N the quantity (6) is a finite-dimensional conic distance.

1.3 Orientation

The condition $\theta_\lambda \geq 0$ on $\mathcal{E}$ is a one-sided sign condition on a divisor sum of level Q , of the shape that appears in sieve theory; it is *not* by itself an admissibility condition for a classical upper-bound sieve weight, since the usual normalisation and majorisation requirements are absent. The comparison we draw is formal: $\kappa(N)$ is the extremal value of a finite convex optimisation problem — a nonnegative least-squares problem in the primal form (6), a second-order cone problem in the dual form of Proposition 1 — whose constraints are divisor-sum inequalities together with a Euclidean norm bound, and the classical parity phenomenon [10, ch. 16] concerns a different object, namely what main terms a sieve can produce without resolving the parity of ω .

1.4 Results

Theorem 2. *For every fixed $A > 0$,*

$$\kappa(N) \ll_A (\log Q)^{-A}.$$

In particular $\kappa(N) \rightarrow 0$: by (8) the optimal nonnegative reweighting beats the uniform baseline by more than any fixed power of $\log Q$.

Theorem 2 is the main result. It comes from one quantitative estimate carrying a parameter β , fixed before N grows, which we state next; Theorem 2 follows by letting β grow with A .

Theorem 3. *Let $\beta \geq 4$ be fixed and put $Y = L^\beta$. For every $\varepsilon > 0$ there is an $N_0(\beta, \varepsilon)$ such that for every integer $N \geq N_0(\beta, \varepsilon)$, with $Q = \lfloor \sqrt{N} \rfloor$ and $L = \log Q$,*

$$\kappa(N) \leq \left(\frac{4}{\sqrt{\beta}} \sqrt{\frac{15}{\pi^2}} + \varepsilon \right) \frac{L^{2-\beta/2}}{\sqrt{\log L}} + (4\sqrt{2e^{e-1}} + \varepsilon) L^2 \exp\left(-\frac{L}{4\beta \log L}\right).$$

All constants and thresholds here may depend on β , which is fixed before N grows; no statement is claimed uniformly in β .

The first term decides the rate, the second is smaller than every power of $1/L$ but carries a large constant. Theorem 2 is the case $\beta = 2A + 4$. Taking instead $\beta = 6$ evaluates both leading coefficients, giving a two-term bound with numerical constants and an unevaluated threshold of validity $N_0(6, \varepsilon)$ (§9); we record it as a concrete example and not as an optimal constant.

Corollary 4. *Put*

$$c_1 = \frac{4}{\sqrt{6}} \sqrt{\frac{15}{\pi^2}} = 2.0131684\dots, \quad c_2 = 4\sqrt{2e^{e-1}} = 13.3565762\dots$$

For every $\varepsilon > 0$ and all sufficiently large N ,

$$\kappa(N) \leq \frac{c_1 + \varepsilon}{L\sqrt{\log L}} + (c_2 + \varepsilon) L^2 \exp\left(-\frac{L}{24\log L}\right) \quad \text{and} \quad \kappa(N) \leq \frac{2.01317 + \varepsilon}{\log Q \sqrt{\log \log Q}}.$$

The displayed decimal 2.01317 exceeds c_1 , as an upper bound requires; c_1 comes from the choice $\beta = 6$. The threshold in Theorem 3 merges those of Proposition 8, Theorem 9 and Theorem 21; see §9 for what is and is not known about it.

Scope: what is proved, and what is not. Both halves are stated here once and are not repeated elsewhere.

What is proved is a statement about one explicit finite convex problem. On the level- Q incidence cone K_Q of (6), nonnegative reweighting of the even- ω columns beats the uniform baseline (8) by more than any fixed power of $\log Q$ (Theorem 2); the weights that do it are explicit (10), grouping the columns by small-prime part and reproducing the retained classes of every Y -smooth row exactly; and the bound is reduced to, and then proved from, one statement about Möbius sums over friable integers. And at seven computable sizes $\kappa(N)$ is located in certified rational intervals, with a certified lower bound at an eighth (§11).

What is not proved: the paper is not a breakthrough on the parity barrier of sieve theory, and it is not a statement about primes. The parity of ω splits the band and carries the construction and the proof; the parity obstruction of sieve theory is a different statement, about sifted sets and primes rather than about this cone, and no implication in either direction between Theorem 2 and the obstruction cited in §1.3 is claimed or used. Every statement above is an upper bound on $\kappa(N)$; none of them asserts that $\kappa(N) > 0$, and none says anything about whether the rate is optimal. Every bound is proved for *fixed* β , and nothing is claimed uniformly in a growing β (Remark 26). The theorem is effective in principle, but no numerical threshold and no useful finite-size estimate has been evaluated — which is not to say that none exists (§9). §11 reports what is known at computable sizes, and none of it bears on Theorem 2.

1.5 Method

The proof exhibits a feasible point of K_Q close to t . Fix a parameter $\beta \geq 4$ and put

$$Y = L^\beta, \quad a_Y(n) = \prod_{p|n, p \leq Y} p, \tag{9}$$

taking $\beta = 6$ only where a numerical constant is wanted (Corollary 4). Group the band by the *small-prime part* a_Y : for squarefree Y -smooth a let B_a and R_a be the number of elements of $\mathcal{E}$ and of $\mathcal{O}$ with $a_Y = a$, and set

$$x_\ell = \frac{R_{a_Y(\ell)}}{R B_{a_Y(\ell)}} \quad \text{if } a_Y(\ell) \leq \sqrt{Q} \text{ and } B_{a_Y(\ell)} > 0; \quad x_\ell = 0 \quad \text{otherwise.} \tag{10}$$

Then $x \geq 0$, so $Ax \in K_Q$ and $\text{dist}_2(t, K_Q) \leq \|t - Ax\|_2$. On every Y -smooth row these weights reproduce the contribution of the retained classes exactly, and the error left on that row is exactly the contribution of the discarded odd classes (Lemma 10); that error is smaller than every power of $1/L$, by a Rankin argument (Lemma 14); and the rows with a prime factor exceeding Y contribute $O((Y \log Y)^{-1/2})$, because both t_d and $(Ax)_d$ are $O(1/d)$ there (Lemma 13, with the prime-sum estimate supplied independently by Lemma 12). That is Theorem 9, whose single arithmetic input is

Hypothesis 5 (PC). There is an absolute C with $R_a \leq C B_a$ for every squarefree Y -smooth $a \leq \sqrt{Q}$.

Hypothesis 5 says that after fixing the small-prime part, the two ω -parities of the rough cofactor occur in bounded ratio. We prove it with $C = 1 + O((\log L)^{-2})$ (Theorem 21). Writing $n = ab$ with $a = a_Y(n)$, the cofactor b runs over squarefree integers with all prime factors in $(Y, Q]$ and $b \leq N/a \in [Q^{3/2}, N]$, and the parity split of such b is governed by

$$U(X) = \#\{b \leq X : \mu^2(b) = 1, Y < P^-(b), P^+(b) \leq Q\}, \quad M_{Y,Q}(X) = \sum_{\substack{b \leq X \\ Y < P^-(b), P^+(b) \leq Q}} \mu(b). \quad (11)$$

Hypothesis 5 follows from $U(X) \gg XV(Y)$ together with $|M_{Y,Q}(X)| = o(U(X))$, where $V(y) = \prod_{p \leq y} (1 - 1/p)$. The unsigned lower bound is a one-step Buchstab peeling against a sieve density valid on a long range (Lemmas 15 and 16).

For the signed estimate we use the following elementary fact: since $Q = \lfloor \sqrt{N} \rfloor$,

$$N < (Q + 1)^2 \leq q_1^2,$$

so an integer $n \leq N$ has at most one prime factor exceeding Q , and that prime divides n to the first power. The large-prime removal is a classical splitting step; what the inequality buys at this particular level is that the removal is exact in one pass and that the cofactor needs *no* upper cutoff, since for $p > Q$ it is automatically smaller than q_1 . Thus (Lemma 18)

$$M_{Y,Q}(X) = M_Y(X) + \sum_{Q < p \leq X} M_Y(X/p), \quad M_Y(z) = \sum_{b \leq z, P^-(b) > Y} \mu(b),$$

with the inner sums unrestricted above, and from there nothing is required beyond the quantitative prime number theorem (P) and the matching Möbius bound (L) of §3.1 (Lemma 19 and Theorem 20). No two-cutoff signed theorem has to be quoted.

Finally, κ carries $\|p\|_2$ in its denominator, so an upper bound for κ needs a *lower* bound for $\|p\|_2$. We prove $\|p\|_2 \geq |p_1| = (1 + o(1))N/(4RL^2)$ in §7, by the same large-prime removal at $d = 1$ followed by an exact integral transform; the main term comes from $\sum_{n \geq 1} \mu(n) \log n/n = -1$. That section also proves $R \gg N$, which the construction needs as well. Both statements are instances of known asymptotics for Möbius sums over friable integers (§2); we give short self-contained proofs in the single case needed.

2 Positioning

2.1 Programming formulations of sieve weights

The feasible set of Proposition 1 is a parity-relaxed version of the sign constraint in the linear- and semidefinite-programming formulations of the sieve. Brady [4, 5] is the principal comparison: the thesis formulates the choice of combinatorial sieve weights as a linear program with pointwise divisor-sum sign constraints and studies its duality ([4, §3.1, §4.1, §7.1]) and its optima in a model problem ([4, §4.2]), computes numerical optima at finite level ([4, §8.1]) and reproduces Selberg's parity example with quantitative two-term asymptotics ([4, §8.2]), while [5] lifts the framework to semidefinite programming over matrix cones and shows that verifying divisor-sum sign constraints is co-NP-hard ([5, §2, Prop. 1]), stating that it proves no new sieve bound. Our constraint set differs in two ways: the sign condition is imposed on the even- ω part of the band only, and the duality is over the cone generated by divisor-incidence columns rather than over partition cones of positive semidefinite matrices. We prove an asymptotic statement about an explicitly listed family of finite constraints and make no complexity claim; nothing here bears on [5], whose input representation is different. Related formulations

appear in Heath-Brown [11, §4], where divisor-sum weights are presented as a linear program with pointwise inequalities and a dual read off from extremal sequences graded by the parity of Ω , and in Ford–Maynard [7, Thms. 7.3–7.4]. Rigorous numerical work on sieve constants is established practice ([13, §7.1], [3, §§5.1–5.2], and, with proved approximation-error bounds rather than certificates, [8, §§7–8]); we claim no priority for a programming formulation of sieve weights, for conic duality in sieve theory, or for certified sieve computation.

2.2 Möbius sums over friable and over rough integers

The counting lemmas of §5 and the asymptotic of §7 are weak forms of studied objects; we record this so that no novelty is read into them.

- $U(X)$ in (11) is the squarefree analogue of the two-cutoff count $\Theta(x, y, z) = \#\{n \leq x : \text{all prime factors in } (z, y]\}$. That count was first studied by Friedlander [9], who for fixed values of the two ratios of logarithms obtained an asymptotic whose factor is defined by a system of differential-difference equations; sharper estimates are due to Saias [15], continued in [16] (for this ordering see [12, §7]). All of them count unrestricted integers. For the unrestricted analogue, specialising the modulus in [18, (1.4)] to the primorial through Y compares the sifted friable count with $\Psi(x, y)$, whose size is given by [12, Thm. 1.1]; passing to squarefree integers requires a further adjustment. Lemma 16 is a weak consequence.
- Lemma 17 is a weak form of [12, Cor. 1.3], which gives $\Psi(x, y) = x u^{-(1+o(1))u}$ as y and $u = \log x / \log y$ tend to infinity, uniformly for $u \leq y^{1-\varepsilon}$ — a range strictly containing the one used here.
- Lemma 15 is the Buchstab asymptotic restricted to $u \rightarrow \infty$; see [19, ch. III.6].
- The convolution identity behind Lemma 19, and its combination with the prime number theorem, are proved here. Their object — the Möbius sum over integers free of small prime factors — is the subject of Alladi [1], which is cited as a whole for that object and for the tradition this argument belongs to; no locator is printed and no claim is made about where inside that paper a particular method appears. His companion paper [2] treats the Möbius sum over *friable* integers, the object of §7; the two papers are easily conflated.
- The asymptotic $P_1 = -N/(\log N)^2 + O(N/(\log N)^3)$ of §7 is such a friable Möbius sum, at $u = \log N / \log Q \rightarrow 2$. For it in the form used here see de la Bretèche and Tenenbaum [6, Cor. 1.3, (1.23)]: specialising to μ and taking $y = \sqrt{N} + \frac{1}{2}$ puts u below 2, which satisfies the one-sided exclusion condition there, and gives leading term $x \varpi'(u) / \log^2 y$ with $\varpi'(u) = -1/u^2$, i.e. $-N / \log^2 N$; changing the cutoff from y to Q affects at most one prime and costs $O(\sqrt{N})$, as does the lower band edge. Our §7 is an elementary derivation in the single case needed, not a new asymptotic.

2.3 What is new

The construction (10): grouping the even- ω columns by small-prime part and weighting each class by the ratio of its odd to its even count produces a feasible point of K_Q which, on every Y -smooth row, reproduces the contribution of the retained classes exactly, the remaining error being exactly the contribution of the discarded odd classes on that row; this reduces $\kappa(N) \rightarrow 0$ to the arithmetic statement Hypothesis 5. Applying this conditional class reweighting to this particular finite cone is the contribution. The large-prime removal of Lemma 18 is classical; the observation we use is only that at the level $Q = \lfloor \sqrt{N} \rfloor$ it needs no second pass and leaves the cofactor unrestricted above, which is why Theorem 20 needs nothing beyond

the quantitative prime number theorem (P) and the matching Möbius bound (L). We have not located a published form of Theorem 20 in the uniformity used here, and we claim no novelty for it: the surrounding machinery is standard and we have not settled the question.

What the paper does and does not claim is stated once in §1.4 and is not restated here.

3 Notation, inputs, and elementary lemmas

In addition to (1)–(9) we write

$$\theta = \frac{1}{\log Y}, \quad Z_0 = \exp\left(\frac{L}{(\log L)^3}\right), \quad S = \sum_{p \leq Y} \frac{1}{p}, \quad V(y) = \prod_{p \leq y} \left(1 - \frac{1}{p}\right), \quad (12)$$

and, for $z \geq 1$,

$$\Phi(z) = \#\{n \leq z : P^-(n) > Y\}, \quad M_Y(z) = \sum_{b \leq z, P^-(b) > Y} \mu(b), \quad M(x) = \sum_{n \leq x} \mu(n), \quad (13)$$

with $n = 1$ counted in Φ ; also $\Psi(z, y) = \#\{n \leq z : P^+(n) \leq y\}$ and $Q_2(x) = \sum_{n \leq x} \mu^2(n)$, and γ is Euler's constant. Note that $M(x) = M(\lfloor x \rfloor)$ for real x .

Asymptotic convention. The parameter β of (9) is fixed before N grows and is treated throughout as a constant. Every constant implied by O , o and $\ll$ below may therefore depend on β , and on ε wherever an ε appears, and is otherwise absolute; we write a subscript only where we wish to draw attention to the dependence, so its absence records emphasis and not uniformity. In particular no statement below is claimed uniformly in a growing β : letting $\beta \rightarrow \infty$ with N would require the fresh uniformity analysis described in Remark 26, at the end of §8, which we do not carry out. With β fixed, $S \asymp \log \log L$ and $V(Y) \sim e^{-\gamma}/(\beta \log L)$ by Mertens.

Standing assumption. From §4 onward, N is assumed large enough that $R > 0$, $p \neq 0$ and

$$2 \leq Y < Q, \quad (14)$$

which is possible because $Y = (\log Q)^\beta$. Condition (14) is used in Lemma 18 and nowhere stated again.

3.1 Classical inputs

Each is used only in the form stated.

(M1) $\sum_{p \leq x} (\log p)/p = \log x + O(1)$; explicitly $< \log x$ for $x > 1$ [14, (3.24)].

(M2) $\sum_{p \leq x} 1/p = \log \log x + O(1)$; and for $x \geq y \geq 2$, $\sum_{y < p \leq x} 1/p = \log(\log x / \log y) + O(1/\log y)$.

(M3) $V(y) = e^{-\gamma}(\log y)^{-1}(1 + O(1/\log y))$.

(C) Chebyshev: $\pi(x) \ll x/\log x$; explicitly $\pi(y) \leq 1.26 y/\log y$ for $y > 1$, a rounding of the bound $\pi(y) < 1.25506 y/\log y$ of [14, (3.6)].

(P) There are absolute effective $C_\pi, c_\pi > 0$ with $|\pi(y) - \text{li}(y)| \leq C_\pi y \exp(-c_\pi \sqrt{\log y})$ for $y \geq 3$.

(L) There are absolute effective $C_M, c_M > 0$ with $|M(x)| \leq C_M x \exp(-c_M (\log x)^{1/4})$ for real $x \geq 2$.

(M1)–(M3) and (C) are Mertens’ and Chebyshev’s theorems. Inputs (P) and (L) are both consequences, by standard contour integration, of the de la Vallée Poussin zero-free region [20, Thm. 3.8] together with the bounds for ζ'/ζ and $1/\zeta$ on it [20, Thm. 3.11] and truncated Perron inversion [20, Lem. 3.12]; both are quoted here in deliberately weakened shapes. For explicit constants in (P), Theorem 2 of [21] gives

$$|\pi(x) - \text{li}(x)| \leq 0.2795 x(\log x)^{-3/4} \exp\left(-\sqrt{\frac{\log x}{6.455}}\right) \quad (x \geq 229),$$

whose logarithmic factor is at most 1 for $x \geq e$ and whose exponential constant is $1/\sqrt{6.455} = 0.3935\dots$; enlarging C_π over the finite range $3 \leq x < 229$ yields (P) with $c_\pi = 0.39$. Nothing below depends on the value of c_π , which enters only through $e^{-c_\pi\sqrt{L}}$ being smaller than every power of $1/L$. We also use

$$Q_2(x) = \frac{6}{\pi^2}x + O(\sqrt{x}), \quad (15)$$

which follows from $Q_2(x) = \sum_{d \leq \sqrt{x}} \mu(d) \lfloor x/d^2 \rfloor = x \sum_{d \leq \sqrt{x}} \mu(d) d^{-2} + O(\sqrt{x})$ and $\sum_{d > \sqrt{x}} d^{-2} \ll x^{-1/2}$.

Lemma 6. $\sum_{n \geq 1} \frac{\mu(n)}{n} = 0$ and $\sum_{n \geq 1} \frac{\mu(n) \log n}{n} = -1$.

Proof. By partial summation and (L), $\sum_{k > T} \mu(k)/k = -M(T)/T + \int_T^\infty M(v)v^{-2}dv \rightarrow 0$ as $T \rightarrow \infty$, and likewise with $\log k$ inserted; so both series converge. For $\sigma > 1$ we have $\sum_n \mu(n)n^{-\sigma} = \zeta(\sigma)^{-1}$ and, differentiating, $\sum_n \mu(n)(\log n)n^{-\sigma} = \zeta'(\sigma)/\zeta(\sigma)^2$. Since a Dirichlet series converging at $s = 1$ is continuous from the right there (Abel’s theorem for Dirichlet series), the two sums equal $\lim_{\sigma \rightarrow 1+} \zeta(\sigma)^{-1} = 0$ and $\lim_{\sigma \rightarrow 1+} \zeta'(\sigma)/\zeta(\sigma)^2 = -1$, the latter from $\zeta(\sigma) = (\sigma - 1)^{-1} + \gamma + O(\sigma - 1)$ and $\zeta'(\sigma) = -(\sigma - 1)^{-2} + O(1)$. $\square$

Lemma 7. For every integer $N \geq 4$:

(i) $Q^2 \leq N < (Q + 1)^2 \leq q_1^2$; consequently every $n \leq N$ has at most one prime factor exceeding Q , and that prime divides n exactly once;

(ii) $Q/2 - 1 \leq \text{thr} \leq Q$;

(iii) $2L \leq \log N < 2L + 2/Q$, so $L/\log N \leq 1/2$.

Proof. (i) $Q = \lfloor \sqrt{N} \rfloor$ gives $Q + 1 > \sqrt{N}$, so $N < (Q + 1)^2$; and $q_1 \geq Q + 1$ because q_1 is an integer exceeding Q . Two primes exceeding Q (equal or not) dividing $n \leq N$ would force $n \geq q_1^2 > N$. (ii) $N < (Q + 1)^2 \leq q_1(Q + 1)$ gives $N/q_1 < Q + 1$, so $\text{thr} \leq Q$; and $q_1 \leq 2Q$ by Bertrand gives $\text{thr} \geq N/(2Q) - 1 \geq Q/2 - 1$. (iii) $Q^2 \leq N < (Q + 1)^2$ and $\log(Q + 1) < L + 1/Q$. $\square$

Part (i) is why Theorem 2 concerns every large N and not a subsequence: it replaces the hypothesis $N = Q^2$ in the large-prime removals of Lemmas 16, 18 and 22.

Proof of Proposition 1. K_Q is finitely generated, hence a closed convex cone, so with $K_Q^\circ = \{w : \langle w, v \rangle \leq 0 \ \forall v \in K_Q\}$ Moreau’s decomposition (see e.g. [17, Thm. 1.1]) gives

$$\text{dist}_2(t, K_Q) = \max\{\langle t, w \rangle : w \in K_Q^\circ, \|w\|_2 \leq 1\}.$$

Put $\lambda = -w$; then $\langle w, A_\ell \rangle = \theta_w(\ell) \leq 0$ is exactly $\theta_\lambda(\ell) \geq 0$, and by (4)

$$\langle t, w \rangle = - \sum_d t_d \lambda_d = -\frac{1}{R} \sum_d c_{\mathcal{O}}(d) \lambda_d = -\frac{1}{R} \sum_{n \in \mathcal{O}} \theta_\lambda(n) = -\mathbb{E}_{\mathcal{O}} \theta_\lambda,$$

which is the displayed equality. If $t \in K_Q$ the distance is 0 and $w = 0$ attains it. If $t \notin K_Q$ the distance is positive and is attained at $w = (t - \Pi_{K_Q} t) / \|t - \Pi_{K_Q} t\|_2$. For uniqueness, suppose $w_1 \neq w_2$ both lie in K_Q° with $\|w_i\|_2 \leq 1$ and $\langle t, w_i \rangle = D > 0$. Their midpoint w lies in K_Q° , has $\langle t, w \rangle = D$, and has $\|w\|_2 < 1$ by strict convexity of the Euclidean ball; then $w / \|w\|_2 \in K_Q^\circ$ is feasible with value $D / \|w\|_2 > D$, a contradiction. The same rescaling shows $\|\lambda\|_2 = 1$ at the maximum. $\square$

Finally we record the normalisation, proved in §7.

Proposition 8. *Put $P_1 = \sum_{n \in \mathcal{B}} \mu(n)$, a quantity defined without dividing by R . There are absolute effective constants such that, for $\log N$ large,*

$$P_1 = -\frac{N}{(\log N)^2} \left(1 + O\left(\frac{1}{\log N}\right) \right).$$

Consequently $P_1 < 0$ and $|\mathcal{B}| = \frac{6}{\pi^2}(1 - \log 2)N + O(N/L)$, so $R \geq \frac{1}{2}|\mathcal{B}| \geq 0.09N > 0$; hence $N/R = O(1)$, indeed $N/R \rightarrow \pi^2/(3(1 - \log 2)) = 10.7213228\dots$, and $p_1 = P_1/R$ is well defined and satisfies, for every $\varepsilon \in (0, 1)$ and all $L \geq L_0(\varepsilon)$,

$$\|p\|_2 \geq |p_1| \geq (1 - \varepsilon) \frac{N}{R} \cdot \frac{1}{4L^2} > 0.$$

4 The construction and the reduction

Group the band by small-prime part: for squarefree Y -smooth a put $\mathcal{E}(a) = \{\ell \in \mathcal{E} : a_Y(\ell) = a\}$, $B_a = |\mathcal{E}(a)|$, $\mathcal{O}(a) = \{n \in \mathcal{O} : a_Y(n) = a\}$, $R_a = |\mathcal{O}(a)|$, so that $\mathcal{E} = \bigsqcup_a \mathcal{E}(a)$ and $\mathcal{O} = \bigsqcup_a \mathcal{O}(a)$. With x as in (10) set

$$e_d = t_d - (Ax)_d, \quad \text{so} \quad \text{dist}_2(t, K_Q) \leq \|e\|_2, \quad (16)$$

and

$$\delta = \frac{1}{R} \#\{n \in \mathcal{O} : a_Y(n) > \sqrt{Q}\}. \quad (17)$$

For every *retained* class $a \leq \sqrt{Q}$, Hypothesis 5 implies $B_a = 0 \Rightarrow R_a = 0$, since $R_a \leq C B_a$ at $B_a = 0$ forces $R_a = 0$: that is the whole of the hypothesis, one inequality. It matters here because (10) is defined for a retained class with $B_a = 0 < R_a$, which it weights by zero, but then nothing reproduces that class's odd mass, which survives as the dead-class term of (18) below. Classes with $a > \sqrt{Q}$ are discarded outright and are not constrained by Hypothesis 5. By Theorem 21 no retained class with $B_a = 0 < R_a$ occurs once L is large.

Theorem 9. *Assume Hypothesis 5 with constant C . For every $\varepsilon > 0$ and all $L \geq L_0(\varepsilon)$,*

$$\kappa(N) \leq (4 + \varepsilon)L^2 \left[\sqrt{\frac{15}{\pi^2}} \max(1, C) \left(\sum_{p > Y} p^{-2} \right)^{1/2} + \left(\frac{2\delta R}{N} \right)^{1/2} \right].$$

4.1 The Y -smooth rows

Lemma 10. *Let $d \leq Q$ be squarefree with $P^+(d) \leq Y$. Then, unconditionally,*

$$e_d = \frac{1}{R} \left[\sum_{a > \sqrt{Q}, d|a} R_a + \sum_{a \leq \sqrt{Q}, B_a=0, d|a} R_a \right], \quad (18)$$

and under Hypothesis 5 the second sum vanishes, so

$$e_d = \frac{1}{R} \#\{n \in \mathcal{O} : d \mid n, a_Y(n) > \sqrt{Q}\} \in [0, \min(\delta, t_d)]. \quad (19)$$

In particular, under Hypothesis 5, $e_1 = \delta$: on the row $d = 1$ the error is then exactly the discarded odd mass. Unconditionally, (18) at $d = 1$ adds to that the mass of the retained classes with $B_a = 0$.

Proof. Every prime factor of d is at most Y and d is squarefree, so for squarefree ℓ one has $d \mid \ell$ if and only if $d \mid a_Y(\ell)$. Hence $\#\{\ell \in \mathcal{E}(a) : d \mid \ell\} = B_a \mathbf{1}_{d|a}$ and, by (10),

$$(Ax)_d = \frac{1}{R} \sum_{a \leq \sqrt{Q}, B_a > 0, d|a} R_a, \quad t_d = \frac{1}{R} \sum_{a: d|a} R_a,$$

the second because $\mathcal{O}$ is partitioned by a_Y . Subtracting gives (18): the classes absent from $(Ax)_d$ are those with $a > \sqrt{Q}$ and those with $a \leq \sqrt{Q}$, $B_a = 0$. For a retained class $a \leq \sqrt{Q}$ with $B_a = 0$, Hypothesis 5 forces $R_a \leq C \cdot 0 = 0$, so every term of the second sum is zero and the sum vanishes (its index set need not be empty). For the range in (19): $e_d \geq 0$ termwise; $e_d \leq t_d$ since $(Ax)_d \geq 0$; and $e_d \leq \delta$ since the set counted lies in $\{n \in \mathcal{O} : a_Y(n) > \sqrt{Q}\}$. $\square$

Thus (19), and with it all of Theorem 9, is conditional on Hypothesis 5; no part of the reduction is hypothesis-free.

Lemma 11. Assume Hypothesis 5 and put $c_0 = N/R$. Then $\|e|_{P^+(d) \leq Y}\|_2 \leq \sqrt{2c_0\delta}$.

Proof. If $\delta = 0$ then $e_d = 0$ on every Y -smooth row by (19); so assume $\delta > 0$. Since $c_{\mathcal{O}}(d) \leq \lfloor N/d \rfloor \leq N/d$ we have $t_d \leq c_0/d$, so $0 \leq e_d \leq f(d)$ with $f(w) = \min(\delta, c_0/w)$. As f^2 is nonincreasing on $(0, \infty)$, $f(d)^2 \leq \int_{d-1}^d f(w)^2 dw$, whence

$$\sum_{d \geq 1} f(d)^2 \leq \int_0^\infty \min\left(\delta^2, \frac{c_0^2}{w^2}\right) dw = \int_0^{c_0/\delta} \delta^2 dw + \int_{c_0/\delta}^\infty \frac{c_0^2}{w^2} dw = 2c_0\delta. \quad \square$$

4.2 The rows with a large prime factor

The two ingredients are separated, because only the second uses Hypothesis 5: the prime-sum estimate below is unconditional and is also the one quoted in §8, so no PC-conditional statement is used anywhere in the proof of Hypothesis 5 itself.

Lemma 12. Unconditionally,

$$\sum_{\substack{d \text{ squarefree} \\ P^+(d) > Y}} d^{-2} \leq \frac{\zeta(2)}{\zeta(4)} \sum_{p > Y} p^{-2}, \quad \sqrt{\frac{\zeta(2)}{\zeta(4)}} = \sqrt{\frac{15}{\pi^2}} = 1.2328088\dots,$$

and $\sum_{p > Y} p^{-2} = (1 + o(1))/(Y \log Y)$ as $Y \rightarrow \infty$.

Proof. Every squarefree d with a prime factor exceeding Y is pm with $p > Y$ prime and m squarefree, so

$$\sum_{\substack{d \text{ squarefree} \\ P^+(d) > Y}} d^{-2} \leq \left(\sum_{p > Y} p^{-2} \right) \sum_{m \text{ squarefree}} m^{-2} = \left(\sum_{p > Y} p^{-2} \right) \prod_p (1 + p^{-2}) = \frac{\zeta(2)}{\zeta(4)} \sum_{p > Y} p^{-2},$$

because $1 + p^{-2} = (1 - p^{-4})/(1 - p^{-2})$; and $\zeta(2)/\zeta(4) = 15/\pi^2$. For the last claim, partial summation with $\pi(t) = (1 + o(1))t/\log t$ gives $\sum_{p > Y} p^{-2} = -\pi(Y)/Y^2 + 2 \int_Y^\infty \pi(t)t^{-3} dt = (1 + o(1))(-1 + 2)/(Y \log Y)$. $\square$

Lemma 13. *Assume Hypothesis 5. Then*

$$\|e|_{P^+(d)>Y}\|_2 \leq \max(1, C) \frac{N}{R} \sqrt{\frac{\zeta(2)}{\zeta(4)}} \left(\sum_{p>Y} p^{-2} \right)^{1/2}.$$

Proof. Let $d \leq Q$ be squarefree with a prime factor exceeding Y . Then $t_d = c_{\mathcal{O}}(d)/R \leq (N/R)/d$; and by (10) with Hypothesis 5 one has $x_\ell \leq C/R$ for every ℓ , so $0 \leq (Ax)_d \leq (C/R) \#\{\ell \in \mathcal{E} : d \mid \ell\} = Cc_{\mathcal{E}}(d)/R \leq C(N/R)/d$. Hence $|e_d| \leq \max(1, C)(N/R)/d$. The factor N/R here is exact and is carried as such: here it enters linearly (it also enters Lemma 11, under a square root), and it cancels in §8 against the R/N that comes from $1/\|p\|_2$, so the density bound of Proposition 8 is not used at this step (Remark 25). Summing the squares over those d and applying Lemma 12 gives the claim. $\square$

4.3 The discarded mass

Lemma 14. *With $\theta = 1/\log Y$ and $\Xi = e^{e-1} = 5.5749415\dots$,*

$$\sum_{n \leq N} a_Y(n)^\theta \leq \Xi N, \quad \text{hence} \quad \delta \leq \Xi \frac{N}{R} \exp\left(-\frac{L}{2 \log Y}\right).$$

Proof. Since $a_Y(n)$ is squarefree and Y -smooth, for $\theta > 0$

$$a_Y(n)^\theta = \prod_{p|n, p \leq Y} (1 + (p^\theta - 1)) = \sum_{d|a_Y(n)} \prod_{p|d} (p^\theta - 1) = \sum_{\substack{d|n, \mu^2(d)=1 \\ P^+(d) \leq Y}} \prod_{p|d} (p^\theta - 1),$$

because $d \mid a_Y(n)$ is the same as $d \mid n$ with d squarefree and Y -smooth. All coefficients are nonnegative. Summing over $n \leq N$ and using $\lfloor N/d \rfloor \leq N/d$,

$$\sum_{n \leq N} a_Y(n)^\theta \leq N \prod_{p \leq Y} \left(1 + \frac{p^\theta - 1}{p}\right).$$

For $p \leq Y$ we have $0 \leq \theta \log p \leq 1$, and $e^v - 1 \leq (e - 1)v$ on $[0, 1]$ by convexity, so by (M1)

$$\sum_{p \leq Y} \frac{p^\theta - 1}{p} \leq (e - 1)\theta \sum_{p \leq Y} \frac{\log p}{p} \leq (e - 1)\theta \log Y = e - 1;$$

now use $\prod(1+u_p) \leq \exp(\sum u_p)$. For the second display, Markov's inequality on $\mathcal{O}$ with threshold $\sqrt{Q}$ gives $\delta \leq R^{-1}Q^{-\theta/2} \sum_{n \in \mathcal{O}} a_Y(n)^\theta \leq (N/R)\Xi Q^{-\theta/2}$, and $Q^{-\theta/2} = \exp(-L/(2 \log Y))$. $\square$

The divisor expansion in Lemma 14 needs no restriction on the range of a , no sieve bound and no splitting of the a -sum.

4.4 Proof of Theorem 9

By (16) and Lemmas 11 and 13,

$$\text{dist}_2(t, K_Q) \leq \sqrt{2(N/R)\delta} + \max(1, C) \frac{N}{R} \sqrt{\frac{15}{\pi^2}} \left(\sum_{p>Y} p^{-2} \right)^{1/2}.$$

By Proposition 8, $1/\|p\|_2 \leq (4+\varepsilon)L^2(R/N)$ for $L \geq L_0(\varepsilon)$. Multiplying, the factor N/R cancels exactly in the second term and halves in the first. $\square$

5 The counting lemmas

The range that matters is $X \in [Q^{3/2}, N]$, i.e. $\log X \in [1.5L, 2L + O(1/Q)]$ by Lemma 7(iii). A single $Z_0 = \exp(L/(\log L)^3)$ serves every lemma below.

Lemma 15. *Uniformly for $z \geq Z_0$, $\Phi(z) = zV(Y)(1 + o(1))$.*

Proof. Let $P = \prod_{p \leq Y} p$, $J = \lceil \log L \rceil$, and for $n \geq 1$ put $k = \omega(\gcd(n, P))$. Legendre's identity reads $\mathbf{1}_{P-(n) > Y} = \sum_{d | \gcd(n, P)} \mu(d)$, and

$$\sum_{j=0}^J (-1)^j \binom{k}{j} = (-1)^J \binom{k-1}{J} \quad (k \geq 1), \quad = 1 \quad (k = 0),$$

so the truncation error at each n is $\binom{k-1}{J} \leq \binom{k}{J}$. Summing over $n \leq z$ and using $\sum_{n \leq z} \binom{\omega(\gcd(n, P))}{J} = \sum_{d | P, \omega(d)=J} \lfloor z/d \rfloor$,

$$\left| \Phi(z) - \sum_{d | P, \omega(d) \leq J} \mu(d) \lfloor z/d \rfloor \right| \leq z \sum_{d | P, \omega(d)=J} \frac{1}{d} \leq \frac{zS^J}{J!}, \quad (20)$$

the last step because the elementary symmetric function obeys $e_J(1/p : p \leq Y) \leq S^J/J!$. Replacing $\lfloor z/d \rfloor$ by z/d and completing the Möbius sum,

$$\sum_{d | P, \omega(d) \leq J} \mu(d) \lfloor z/d \rfloor = zV(Y) - z \sum_{d | P, \omega(d) > J} \frac{\mu(d)}{d} + O(\#\{d | P : \omega(d) \leq J\}), \quad (21)$$

$$\left| \sum_{d | P, \omega(d) > J} \frac{\mu(d)}{d} \right| \leq \sum_{j > J} \frac{S^j}{j!} \leq \frac{S^{J+1}}{(J+1)!} \left(1 - \frac{S}{J+2}\right)^{-1} \leq \frac{2S^{J+1}}{(J+1)!} \quad \text{if } S \leq \frac{J+2}{2}, \quad (22)$$

$$\#\{d | P : \omega(d) \leq J\} \leq (J+1)\pi(Y)^J \leq Y^J. \quad (23)$$

The side condition in (22) holds eventually because $S \asymp \log \log L$ and $J \asymp \log L$; the second inequality of (23) holds eventually because (C) gives $Y/\pi(Y) \gg \log Y$, so $(Y/\pi(Y))^J \geq J+1$. Divide the three errors by $zV(Y)$, which is $\asymp z/(\beta \log L)$ by (M3). Since $S = (1 + o(1)) \log \log L$ and $J = \lceil \log L \rceil$,

$$\frac{S^J}{J!} \leq \left(\frac{eS}{J}\right)^J = \exp(-(\log L)(\log \log L - \log \log \log L - 1 + o(1))),$$

smaller than every fixed power of $1/L$; so (20) and (22) are $o(V(Y))$ with room. For (23) we compare logarithms: $\log(Y^J) = \beta J \log L = \beta(\log L)^2(1 + o(1))$, whereas

$$\log(zV(Y)) \geq \log(Z_0 V(Y)) = \frac{L}{(\log L)^3} - \log(\beta \log L) - \gamma + o(1),$$

and $\beta(\log L)^2 = o(L/(\log L)^3)$ because $(\log L)^5 = o(L)$. All three errors are therefore $o(1)$ relative to $zV(Y)$, uniformly in $z \geq Z_0$. $\square$

Lemma 15 asks only for $z \geq Z_0$, where $u = \log z / \log Y \geq L/(\beta(\log L)^4) \rightarrow \infty$; this avoids the Buchstab transition rather than estimating it. For comparison, with ϖ Buchstab's function one has, uniformly for $2 \leq Y \leq \sqrt{z}$,

$$\Phi(z) = \frac{z}{\log Y} \left\{ \varpi(u) + O\left(\frac{1}{\log Y}\right) \right\}$$

[19, ch. III.6], hence $\Phi(z) \sim \varpi(u)e^\gamma zV(Y)$ as $Y \rightarrow \infty$, uniformly in that range — which does contain the range of Lemma 15, since $2\beta(\log L)^4 \leq L$ eventually. Here $\max_{2 \leq u \leq 3} \varpi(u)e^\gamma =$

1.0101232..., attained at $u = 2.7632228...$, and the maximum can be read off the delay equation: on $[2, 3]$ one has $u\varpi(u) = 1 + \log(u-1)$, so $u^2\varpi'(u) = u/(u-1) - 1 - \log(u-1)$, which written in $v = u-1$ is $1/v - \log v$, strictly decreasing on $[1, 2]$ from 1 to $-0.1931...$ and therefore zero exactly once; there $v \log v = 1$ and $\varpi(u) = 1/v = 0.5671432...$. We make no claim about where the maximum over all $u \geq 2$ lies, because nothing below needs one. What is used is only the limit $\varpi(u)e^\gamma \rightarrow 1$ as $u \rightarrow \infty$, which yields the constant $1 - \log 2$ in Lemma 16.

Lemma 16. *Uniformly for $X \in [Q^{3/2}, N]$,*

$$U(X) \geq (1 - \log 2 + o(1)) X V(Y), \quad 1 - \log 2 = 0.3068528...,$$

and in particular $U(X) \gg X/\log Y$.

Proof. Let $\Phi_Q(z) = \#\{n \leq z : Y < P^-(n), P^+(n) \leq Q\}$. If $n \leq X \leq N$ has $P^-(n) > Y$ and $P^+(n) > Q$, then by Lemma 7(i) the large prime p is unique and divides n once; writing $n = pm$, Lemma 7(i)–(ii) give $m \leq X/p \leq N/q_1 < q_1$, so every prime factor of m is smaller than q_1 , i.e. at most Q — the upper cutoff on m is automatic — and $P^-(m) > Y$ or $m = 1$. The map $n \mapsto (p, m)$ is a bijection onto the pairs counted, so

$$\Phi(X) = \Phi_Q(X) + \sum_{Q < p \leq X} \Phi_Q(X/p), \quad \text{hence} \quad \Phi_Q(X) \geq \Phi(X) - \sum_{Q < p \leq X} \Phi(X/p). \quad (24)$$

Split the subtracted sum at $X/p = Z_0$, using Lemma 15 for $p \leq X/Z_0$ and $\Phi(z) \leq z$ beyond:

$$\sum_{Q < p \leq X} \Phi(X/p) \leq (1 + o(1)) X V(Y) \sum_{Q < p \leq X} \frac{1}{p} + X \sum_{X/Z_0 < p \leq X} \frac{1}{p}.$$

By (M2) and Lemma 7(iii), $\sum_{Q < p \leq X} 1/p = \log(\log X/\log Q) + O(1/L) \leq \log 2 + o(1)$, uniformly for $X \leq N$. For the short block put $\tau_X = \log Z_0/\log X = L/((\log L)^3 \log X)$; then

$$\sum_{X/Z_0 < p \leq X} \frac{1}{p} = -\log(1 - \tau_X) + O(1/L) = \tau_X + O(\tau_X^2 + 1/L) \leq \left(\frac{2}{3} + o(1)\right)(\log L)^{-3}, \quad (25)$$

because $\log X \geq 1.5L$ gives $\tau_X \leq \frac{2}{3}(\log L)^{-3}$, and $1/L = o((\log L)^{-3})$. Note that (25) is an inequality, not an equality: $\tau_X(\log L)^3$ equals $2/3$ at $X = Q^{3/2}$ and $\frac{1}{2} + O(1/(QL))$ at $X = N$. Since $(\log L)^{-3} = o(V(Y))$ the short block is $o(XV(Y))$, and $\Phi(X) = (1 + o(1))XV(Y)$ by Lemma 15; hence $\Phi_Q(X) \geq (1 - \log 2 + o(1))XV(Y)$. Finally, deleting the non-squarefree b costs at most $\sum_{p > Y} X/p^2 \ll X/(Y \log Y) = o(XV(Y))$ by Lemma 12, which is unconditional. $\square$

Lemma 17. *Let $u = \log z/\log Y$. Uniformly for $Z_0 \leq z \leq N$, so that $u \geq L/(\beta(\log L)^4) \rightarrow \infty$,*

$$\Psi(z, Y) \leq z \exp\left(-\frac{1}{2}u \log u\right).$$

Proof. Rankin's trick: for $\alpha \in (0, 1]$ one has $\Psi(z, Y) \leq z^\alpha \prod_{p \leq Y} (1 - p^{-\alpha})^{-1}$, since $\sum_{P^+(n) \leq Y} (z/n)^\alpha \geq \Psi(z, Y)$ termwise. Take $\alpha = 1 - (\log u)/(\log Y)$, so that $z^\alpha = z \exp(-u \log u)$ exactly. Throughout the range $\log u = (1 + o(1)) \log L$ and $\log Y = \beta \log L$, so $1 - \alpha = (1 + o(1))/\beta$ and $\alpha \rightarrow 1 - 1/\beta \in [3/4, 1)$, uniformly. The bound $\alpha \geq 3/4$ is moreover an inequality and not only a limit, for L beyond an absolute point: $z \leq N$ gives $\log z < 2L + 2/Q \leq 3L$, so $u = \log z/\log Y \leq 3L/(\beta \log L) \leq L$, whence $\log u \leq \log L$ and

$$\alpha = 1 - \frac{\log u}{\beta \log L} \geq 1 - \frac{1}{\beta} \geq \frac{3}{4}$$

since $\beta \geq 4$. Estimate $\sum_{p \leq Y} p^{-\alpha}$ by Abel summation:

$$\sum_{p \leq Y} p^{-\alpha} = Y^{-\alpha} \pi(Y) + \alpha \int_2^Y \pi(t) t^{-\alpha-1} dt.$$

Since $Y^{1-\alpha} = u$ and $(1-\alpha) \log Y = \log u$, the boundary term is, by (P),

$$Y^{-\alpha} \pi(Y) = (1+o(1)) \frac{Y^{1-\alpha}}{\log Y} = (1+o(1))(1-\alpha) \frac{u}{\log u} = \left(\frac{1}{\beta} + o(1)\right) \frac{u}{\log u},$$

so it is a definite fraction of the total and not negligible. In the integral, the part $t \leq \sqrt{Y}$ is at most $\int_2^{\sqrt{Y}} t^{-\alpha} dt \leq Y^{(1-\alpha)/2} / (1-\alpha) = O(u^{1/2} \log L) = o(u/\log u)$; on $[\sqrt{Y}, Y]$, (P) gives $\pi(t) = (1+o(1))t/\log t$ uniformly, and substituting $t = Y^s$, so that $t^{1-\alpha} = u^s$ and $dt/(t \log t) = ds/s$,

$$\alpha \int_{\sqrt{Y}}^Y \pi(t) t^{-\alpha-1} dt = (1+o(1)) \alpha \int_{1/2}^1 u^s \frac{ds}{s} = (1+o(1)) \alpha \frac{u}{\log u} = \left(1 - \frac{1}{\beta} + o(1)\right) \frac{u}{\log u}$$

by Laplace's method, the mass sitting at s near 1, i.e. at p near Y , which is where (P) is strongest. The two contributions sum to $(1+o(1))u/\log u$. For the Euler product, peel off $p = 2$, whose factor contributes $-\log(1 - 2^{-\alpha}) = O(1)$ because $2^{-\alpha} \leq 2^{-3/4} = 0.5946 \dots < 1$; for $p \geq 3$ and $\alpha \geq 3/4$ one has $p^{-\alpha} \leq 3^{-3/4} = 0.4386 \dots < 1/2$, so $-\log(1 - v) \leq v + v^2$ applies on its stated range, and $\sum_{3 \leq p \leq Y} p^{-2\alpha} \leq \sum_p p^{-3/2} = O(1)$. Hence $\log \prod_{p \leq Y} (1 - p^{-\alpha})^{-1} = (1+o(1))u/\log u + O(1)$, and $\log(\Psi(z, Y)/z) \leq -u \log u + (1+o(1))u/\log u + O(1) \leq -\frac{1}{2}u \log u$, since $u/\log u = o(u \log u)$. $\square$

6 The parity hypothesis is a theorem

6.1 The parity split

Fix a squarefree Y -smooth $a \leq \sqrt{Q}$. Every $n \in \mathcal{B}$ with $a_Y(n) = a$ is $n = ab$ with b squarefree, $Y < P^-(b)$, $P^+(b) \leq Q$ and $b \in (\text{thr}/a, N/a]$; coprimality of a and b is automatic from $P^-(b) > Y \geq P^+(a)$, and $\omega(n) = \omega(a) + \omega(b)$. With

$$U_a = U(N/a) - U(\text{thr}/a), \quad M_a = M_{Y,Q}(N/a) - M_{Y,Q}(\text{thr}/a)$$

in the notation (11), and $\mu(b) = (-1)^{\omega(b)}$ for squarefree b ,

$$\{B_a, R_a\} = \left\{ \frac{U_a + M_a}{2}, \frac{U_a - M_a}{2} \right\}, \quad \frac{R_a}{B_a} \leq \frac{U_a + |M_a|}{U_a - |M_a|} \quad \text{if } |M_a| < U_a, \quad (26)$$

which of the two is which being decided by the parity of $\omega(a)$. So Hypothesis 5 needs only $|M_a|$ bounded away from U_a by a fixed factor; we prove the stronger $|M_a| = o(U_a)$ because it is available.

6.2 The signed estimate

Lemma 18. *Assume (14). Then for every $X \leq N$,*

$$M_{Y,Q}(X) = M_Y(X) + \sum_{Q < p \leq X} M_Y(X/p).$$

Proof. Split $M_Y(X)$ according to whether $P^+(b) \leq Q$. If not, $P^+(b) = p > Q$; by Lemma 7(i) p is the only prime factor of b exceeding Q and divides b once, so $b = pb'$ with $b' = b/p \leq X/p \leq N/q_1 < q_1$, whence every prime factor of b' is at most Q , $p \nmid b'$, and $P^-(b') > Y$ or $b' = 1$.

Conversely, let $Q < p \leq X$ and $b' \leq X/p$ with $P^-(b') > Y$, and put $b = pb'$. Then $P^+(b') \leq p$ automatically, and $P^-(b) = \min(p, P^-(b')) > Y$ because $p > Q \geq Y$ by (14); so b is counted by $M_Y(X)$ with $P^+(b) > Q$, and the correspondence is a bijection. Since $\mu(pb') = -\mu(b')$, we get $M_Y(X) = M_{Y,Q}(X) - \sum_{Q < p \leq X} M_Y(X/p)$. $\square$

The hypothesis $Y \leq Q$ is needed: without it the converse direction fails. At $N = 256$, $Q = 16$, $Y = (\log 16)^6 > 256$ and $X = 17$, both sides of the identity would read 1 and 2 respectively, the discrepancy being the pair $(p, b') = (17, 1)$, whose product 17 is not counted by $M_Y(17)$.

Lemma 19. *Uniformly for $Z_0 \leq z \leq N$ and for every fixed $A_0 > 0$, $|M_Y(z)| \ll_{A_0, \beta} z(\log L)^{-A_0}$.*

Proof. Every n factors uniquely as $n = mb$ with m Y -friable and $P^-(b) > Y$, and $\mu(n) = \mu(m)\mu(b)$; in Dirichlet series, $\sum_{P^-(b) > Y} \mu(b)b^{-s} = \prod_{p > Y} (1 - p^{-s}) = \zeta(s)^{-1} \prod_{p \leq Y} (1 - p^{-s})^{-1}$, so $M_Y(z) = \sum_{m \leq z, P^+(m) \leq Y} M(z/m)$. Split at $m \leq z/W$ with $W = \exp(\sqrt{\log z})$. For $m \leq z/W$, apply (L) at argument at least W and use $\sum_{m \text{ } Y\text{-friable}} 1/m = V(Y)^{-1} \asymp e^\gamma \log Y$ from (M3):

$$\sum_{m \leq z/W} |M(z/m)| \ll z \exp(-c_M(\log W)^{1/4}) V(Y)^{-1} \ll z(\log L) \exp(-c_M(\log z)^{1/8}).$$

For $m > z/W$ use $|M(z/m)| \leq z/m \leq W$ and Lemma 17: that block is at most $W\Psi(z, Y) \leq z \exp(\sqrt{\log z} - \frac{1}{2}u \log u)$. In the stated range $\log u = (1 + o(1)) \log L$, so

$$u \log u = \frac{\log z}{\beta \log L} (1 + o(1)) \log L = (1 + o(1)) \frac{\log z}{\beta},$$

and $\sqrt{\log z} = o(\log z)$; hence $\sqrt{\log z} - \frac{1}{2}u \log u \leq -(\log z)/(3\beta)$ for L large, uniformly. Since $\log z \geq L/(\log L)^3$, both blocks are $\ll z \exp(-c'L^{1/9})$ with c' depending on β , which beats $z(\log L)^{-A_0}$ for every fixed A_0 . $\square$

Theorem 20. *Uniformly for $X \in [Q^{3/2}, N]$,*

$$|M_{Y,Q}(X)| \leq \left(\frac{2}{3} + o(1)\right) X(\log L)^{-3}.$$

Proof. Apply Lemma 18 and split the prime sum at $X/p = Z_0$. First $|M_Y(X)| \ll X(\log L)^{-4}$ by Lemma 19, as $X \geq Q^{3/2} \gg Z_0$. Next, for $Q < p \leq X/Z_0$, Lemma 19 with $A_0 = 4$ gives $|M_Y(X/p)| \ll (X/p)(\log L)^{-4}$, while by (M2) with Lemma 7(iii) one has $\sum_{Q < p \leq X} 1/p = \log(\log X / \log Q) + O(1/L)$ uniformly, which since $X \leq N$ is at most $\log 2 + o(1)$, the limit at $X = Q^{3/2}$ being $\log(3/2)$; only the resulting $O(1)$ is used, so this block is $\ll X(\log L)^{-4}$. Finally, for $X/Z_0 < p \leq X$ use $|M_Y(z)| \leq z$ and (25), which gives at most $(\frac{2}{3} + o(1))X(\log L)^{-3}$. $\square$

6.3 The hypothesis

Theorem 21. *For all L beyond a point depending only on β , Hypothesis 5 holds with*

$$\frac{R_a}{B_a} \leq 1 + \left(\frac{4\beta e^\gamma}{3(1 - \log 2)} + o(1) \right) (\log L)^{-2} = 1 + O_\beta((\log L)^{-2}),$$

uniformly in squarefree Y -smooth $a \leq \sqrt{Q}$. At $\beta = 6$ the displayed constant is 46.4345.... In particular $B_a > 0$ for every such a . The implied constant is proportional to β , as the displayed coefficient shows; the uniformity asserted is in a , at a fixed β .

Proof. For $a \leq \sqrt{Q}$ we have $X := N/a \in [N/\sqrt{Q}, N] \subseteq [Q^{3/2}, N]$, because $N \geq Q^2$ gives $N/\sqrt{Q} \geq Q^{3/2}$. The lower band edge costs nothing, uniformly: by Lemma 7(ii), $\text{thr}/a \leq Q/a$, so $U(\text{thr}/a) \leq Q/a$, $|M_{Y,Q}(\text{thr}/a)| \leq Q/a$, and $(Q/a)/((N/a)V(Y)) = Q/(NV(Y)) \ll (\log L)/Q \rightarrow 0$. Hence by Lemma 16, Theorem 20 and $V(Y) = (1 + o(1))e^{-\gamma}/(\beta \log L)$ from (M3),

$$U_a \geq (1 - \log 2 + o(1)) \frac{N}{a} V(Y), \quad |M_a| \leq \left(\frac{2}{3} + o(1)\right) \frac{N}{a} (\log L)^{-3},$$

$$r := \frac{|M_a|}{U_a} \leq \frac{(\frac{2}{3} + o(1))(\log L)^{-3}}{(1 - \log 2 + o(1))V(Y)} = \frac{\frac{2}{3}\beta e^\gamma + o(1)}{(1 - \log 2)(\log L)^2}.$$

So $|M_a| < U_a$ for L large, whence $B_a \geq (U_a - |M_a|)/2 > 0$, and (26) gives $R_a/B_a \leq (1+r)/(1-r) = 1 + 2r + O(r^2)$, which is the display. Every estimate depends on a only through $X = N/a$, and Lemma 16 and Theorem 20 are uniform on $[Q^{3/2}, N]$, so no union bound over the classes is required. $\square$

7 The normalisation

This section proves Proposition 8. Write $Y_1 = \text{thr}$, $P_1 = \sum_{n \in \mathcal{B}} \mu(n)$,

$$\Phi_\mu(x) = \sum_{m \leq x, P^+(m) \leq Q} \mu(m), \quad F(y) = \#\{n \leq y : \mu^2(n) = 1, P^+(n) \leq Q\}, \quad \Delta_\pi = \pi - \text{li}.$$

Nothing in this section involves Y or the construction.

Lemma 22. *For every integer $N \geq 4$,*

$$P_1 = M(N) - M(Y_1)(1 + \pi(Q)) + \sum_{k \leq Y_1} \mu(k)\pi(N/k).$$

Proof. $P_1 = \Phi_\mu(N) - \Phi_\mu(Y_1)$, and $\Phi_\mu(Y_1) = M(Y_1)$ because $Y_1 = \text{thr} \leq Q$ by Lemma 7(ii) makes every $m \leq Y_1$ automatically Q -friable. For $\Phi_\mu(N)$: by Lemma 7(i) a squarefree $m \leq N$ has at most one prime factor exceeding Q , and if $m = pm'$ with $p > Q$ then $m' \leq \lfloor N/q_1 \rfloor = Y_1 \leq Q$, so m' is Q -friable, $p \nmid m'$ and $\mu(m) = -\mu(m')$. Summing, $M(N) = \Phi_\mu(N) - \sum_{Q < p \leq N} \Phi_\mu(N/p) = \Phi_\mu(N) - \sum_{Q < p \leq N} M(N/p)$. Finally $\sum_{Q < p \leq N} M(N/p) = \sum_k \mu(k) \#\{p : Q < p \leq N/k\} = \sum_{k \leq Y_1} \mu(k)[\pi(N/k) - \pi(Q)]$, the terms with $k > Y_1$ vanishing because then $N/k < q_1$ and no prime lies in (Q, q_1) ; and $\sum_{k \leq Y_1} \mu(k) = M(Y_1)$. $\square$

Lemma 23. *With $T = \sum_{k \leq Y_1} \mu(k)\text{li}(N/k)$,*

$$T = M(Y_1)\text{li}(N/Y_1) + N \int_1^{Y_1} \frac{M(v)}{v^2 \log(N/v)} dv.$$

Proof. Write $\text{li}(N/k) = \text{li}(2) + \text{Li}(N/k)$ with $\text{Li}(y) = \int_2^y dt/\log t$, legitimate since $N/k \geq N/Y_1 \geq q_1 > Q \geq 2$. Then

$$\sum_{k \leq Y_1} \mu(k) \int_2^{N/k} \frac{dt}{\log t} = \int_2^N \frac{M(\min(Y_1, N/t))}{\log t} dt = M(Y_1)\text{Li}(N/Y_1) + \int_{N/Y_1}^N \frac{M(N/t)}{\log t} dt,$$

the split being legitimate because $N/Y_1 \geq q_1 > 2$; and $v = N/t$ turns the last integral into $N \int_1^{Y_1} M(v)v^{-2} \log(N/v)^{-1} dv$. Adding $\text{li}(2)M(Y_1)$ and using $\text{li}(2) + \text{Li}(y) = \text{li}(y)$ gives the statement. $\square$

Lemma 24. *Put $J_i = \int_1^{Y_1} M(v)(\log v)^i v^{-2} dv$ and $m_i = \sum_{k \leq Y_1} \mu(k)(\log k)^i/k$. Then*

- (i) $|J_i| \leq \Lambda_i = O_i(1)$ for every $i \geq 0$, uniformly in Y_1 ;
- (ii) $J_0 = m_0 - M(Y_1)/Y_1$ and $J_1 = m_1 + J_0 - M(Y_1) \log Y_1/Y_1$;
- (iii) $m_0 = O(\eta_0)$ and $m_1 = -1 + O(\eta_1)$, where $\eta_j \ll (\log Y_1)^{(4j+3)/4} \exp(-c_M(\log Y_1)^{1/4})$.

Proof. (i) On $[1, 2)$ use $|M(v)| = |M(\lfloor v \rfloor)| \leq v$, so that

$$\int_1^2 |M(v)| (\log v)^i \frac{dv}{v^2} \leq \int_1^2 (\log v)^i \frac{dv}{v} = \frac{(\log 2)^{i+1}}{i+1}.$$

On $[2, \infty)$ apply (L) at the real argument v , which is legitimate since $M(v) = M(\lfloor v \rfloor)$ and (L) is stated for real $x \geq 2$; then substitute $w = \log v$ and extend the lower limit from $\log 2$ down to 0, so as to obtain

$$\begin{aligned} \int_2^\infty |M(v)| (\log v)^i \frac{dv}{v^2} &\leq C_M \int_{\log 2}^\infty e^{-c_M w^{1/4}} w^i dw \leq C_M \int_0^\infty e^{-c_M w^{1/4}} w^i dw \\ &= 4C_M (4i+3)! c_M^{-(4i+4)}. \end{aligned}$$

(ii) is partial summation over the unit steps of M , with Y_1 an integer: $J_0 = \sum_{k < Y_1} M(k)(1/k - 1/(k+1))$ telescopes to $m_0 - M(Y_1)/Y_1$, and with $a_k = (\log k + 1)/k$, $J_1 = \sum_{k < Y_1} M(k)(a_k - a_{k+1}) = m_1 + m_0 - M(Y_1)(1 + \log Y_1)/Y_1$, which is the stated form. (iii) By Lemma 6 the completed series are 0 and -1 , so m_j differs from its limit by $-\sum_{k > Y_1} \mu(k)(\log k)^j/k$, which partial summation and (L) bound by η_j as displayed. $\square$

Since $\log Y_1 \geq \log(Q/2 - 1) \geq L - \log 4$ for $Q \geq 8$ by Lemma 7(ii), both η_j are $\ll L^{(4j+3)/4} \exp(-c' L^{1/4})$, smaller than every power of $1/L$.

Proof of Proposition 8. By Lemma 22 and $\pi = \text{li} + \Delta_\pi$,

$$P_1 = \underbrace{M(N)}_{\mathcal{R}_1} - \underbrace{M(Y_1)(1 + \pi(Q))}_{\mathcal{R}_2} + \underbrace{\sum_{k \leq Y_1} \mu(k) \Delta_\pi(N/k)}_{\mathcal{R}_3} + T.$$

By (L), $|\mathcal{R}_1| \leq C_M N \exp(-c_M(\log N)^{1/4})$. By (L) and (C),

$$|\mathcal{R}_2| \leq C_M Q \exp(-c_M(\log Y_1)^{1/4}) (1 + 1.26 Q/L) \leq 3C_M N \exp(-c' L^{1/4})/L,$$

using $1 \leq Q/L$ and $Q^2 \leq N$. For $\mathcal{R}_3$, every argument satisfies $N/k \geq N/Y_1 \geq q_1 > Q$, so $\log(N/k) > L$ and (P) gives

$$|\mathcal{R}_3| \leq C_\pi N e^{-c_\pi \sqrt{L}} \sum_{k \leq Y_1} \frac{1}{k} \leq C_\pi N (1 + L) e^{-c_\pi \sqrt{L}}.$$

All three are $\ll N \exp(-c'' L^{1/4})$, hence $o(N/(\log N)^A)$ for every A .

For T , Lemma 23 together with the elementary estimate $\text{li}(y) = (y/\log y)(1 + O(1/\log y))$ gives

$$|M(Y_1) \text{li}(N/Y_1)| \ll \frac{N}{L} \exp(-c'(\log Y_1)^{1/4}),$$

again negligible; we do not use (C) here, which bounds π and not li . For the integral, put $w = \log v / \log N$, which lies in $[0, 1/2]$ by Lemma 7(ii)–(iii), and expand $(1 - w)^{-1} = 1 + w + w^2(1 - w)^{-1}$, the last term here being at most $2w^2$:

$$\int_1^{Y_1} \frac{M(v) dv}{v^2 \log(N/v)} = \frac{1}{\log N} \left[J_0 + \frac{J_1}{\log N} + E \right], \quad |E| \leq \frac{2\Lambda_2}{(\log N)^2}.$$

By Lemma 24(ii)–(iii), $J_0 = O(\eta_0 + C_M e^{-c' L^{1/4}})$ and $J_1 = -1 + O(\eta_1 + \eta_0 + L C_M e^{-c' L^{1/4}})$, so

$$T = -\frac{N}{(\log N)^2} + O\left(\frac{N\eta_0}{\log N}\right) + O\left(\frac{N\eta_1}{(\log N)^2}\right) + O\left(\frac{2N\Lambda_2}{(\log N)^3}\right) + O\left(\frac{N}{L} e^{-c' L^{1/4}}\right).$$

Each error is $O(N/(\log N)^3)$ once $\log N$ exceeds a threshold determined by (C_M, c_M, C_π, c_π) alone, which gives the displayed asymptotic. In particular $P_1 < 0$, and for every $\varepsilon > 0$ one has $|P_1| \geq (1 - \varepsilon)N/(\log N)^2$ for $\log N$ large; with $\log N < 2L + 2/Q$ from Lemma 7(iii) this yields $|p_1| = |P_1|/R \geq (1 - \varepsilon')(N/R)/(4L^2)$ — legitimate because $R > 0$, which the band count below supplies — and $\|p\|_2 \geq |p_1|$ because $d = 1$ is a coordinate.

For the band size, the bijection of Lemma 22 applied to the squarefree indicator gives, exactly, $F(N) = Q_2(N) - \sum_{Q < p \leq N} Q_2(N/p)$. By (15), $Q_2(x) = \frac{6}{\pi^2}x + O(\sqrt{x})$; by (P), $\sum_{Q < p \leq N} 1/p = \log(\log N / \log Q) + O(L^{-2}) = \log 2 + O(L^{-2})$; and by (C) with partial summation, $\sum_{Q < p \leq N} \sqrt{N/p} \ll N/L$. Hence $F(N) = \frac{6}{\pi^2}(1 - \log 2)N + O(N/L)$, and since $F(\text{thr}) \leq \text{thr} \leq Q$ we get $|\mathcal{B}| = \frac{6}{\pi^2}(1 - \log 2)N + O(N/L)$. Finally $|\mathcal{B}| = |\mathcal{E}| + R$ and $P_1 = |\mathcal{E}| - R$, so $R = \frac{1}{2}(|\mathcal{B}| - P_1) \geq \frac{1}{2}|\mathcal{B}|$ once $P_1 \leq 0$; as $\frac{3}{\pi^2}(1 - \log 2) = 0.0932720\dots$, this gives $R \geq 0.09N$ for L large, and $N/R \rightarrow \pi^2/(3(1 - \log 2))$ follows from $P_1 = o(N)$. $\square$

Remark 25. The lower bound is at bottom the nonvanishing of one classical constant: the main term of T comes from $\sum_{n \geq 1} \mu(n) \log n / n = -1$, while $\sum_{n \geq 1} \mu(n)/n = 0$ annihilates the would-be larger term. Only the sign of P_1 is used for $R \gg N$. As recorded in §2.2, the asymptotic itself is not new; we give the derivation because it is short in the single case needed and keeps the normalisation on the same elementary footing as the rest of the proof.

It is worth saying how much of this the final bound actually consumes. The quantitative density $N/R = O(1)$ is what makes the *absolute* row-error statements of §4 meaningful, $t_d \leq (N/R)/d$ and $e_d \leq \min(\delta, (N/R)/d)$ being bounds on quantities carrying the factor $1/R$. In the *relative* estimate it largely cancels: once Rankin is inserted, each of the two terms of §8 acquires a factor N/R from the numerator and a factor R/N from $1/\|p\|_2$, so what survives of Proposition 8 at the end is the lower bound on $\|p\|_2$ and not the density. Nor is the density needed for that lower bound: the inequality $|p_1| \geq (1 - \varepsilon)N/(4RL^2)$ is the asymptotic for P_1 divided by R , and needs only $R > 0$. So $N/R = O(1)$ is information about the band rather than a step the final estimate cannot do without; we record it because the absolute row bounds are stated in those terms and because the limiting value is worth knowing, not because the argument turns on it.

8 Proof of Theorems 2 and 3

Let $\beta \geq 4$ be fixed. By Theorem 21, Hypothesis 5 holds with $\max(1, C) = 1 + o(1)$. Insert into Theorem 9 the estimate $\sum_{p > Y} p^{-2} = (1 + o(1))/(Y \log Y) = (1 + o(1))/(\beta L^\beta \log L)$ of Lemma 12, and the bound $\delta R/N \leq \Xi \exp(-L/(2\beta \log L))$ of Lemma 14, noting that N/R cancels in the latter as well. The first term of Theorem 9 becomes

$$(4 + \varepsilon)L^2 \sqrt{\frac{15}{\pi^2}} \cdot \frac{1 + o(1)}{\sqrt{\beta L^\beta \log L}} = (1 + o(1)) \frac{4 + \varepsilon}{\sqrt{\beta}} \sqrt{\frac{15}{\pi^2}} \cdot \frac{L^{2-\beta/2}}{\sqrt{\log L}}, \quad (27)$$

and the second becomes

$$(4 + \varepsilon)L^2 \left(2\Xi \exp\left(-\frac{L}{2\beta \log L}\right)\right)^{1/2} = (4 + \varepsilon)\sqrt{2e^{e-1}} L^2 \exp\left(-\frac{L}{4\beta \log L}\right), \quad (28)$$

an equality rather than an asymptotic, because $\Xi = e^{e-1}$ exactly.

Together, (27) and (28) are Theorem 3: absorbing the $(1 + o(1))$ of (27) into an arbitrary $\varepsilon > 0$ gives the two displayed coefficients $4\beta^{-1/2}\sqrt{15/\pi^2} + \varepsilon$ and $4\sqrt{2e^{e-1}} + \varepsilon$, and every $o(1)$ and every threshold used along the way depends only on β and ε , both fixed before N grows. $\square$

For Theorem 2, fix $A > 0$ and apply Theorem 3 with $\beta = 2A + 4$. The first term is $\ll_A L^{2-\beta/2}(\log L)^{-1/2} = L^{-A}(\log L)^{-1/2} \leq L^{-A}$, and the second is $\ll_A L^{-A}$ because $\exp(-L/((8A+16)\log L))$ decays faster than every power of L . Hence $\kappa(N) \ll_A L^{-A} = (\log Q)^{-A}$. $\square$

For Corollary 4, take $\beta = 6$: then (27) is $(1 + o(1))(4 + \varepsilon)\sqrt{15/\pi^2}/\sqrt{6} \cdot (L\sqrt{\log L})^{-1}$ and (28) is $(4 + \varepsilon)\sqrt{2e^{e-1}}L^2 \exp(-L/(24\log L))$. Since $\varepsilon > 0$ was arbitrary and $4\sqrt{15/\pi^2}/\sqrt{6} = c_1$, $4\sqrt{2e^{e-1}} = c_2$, the first stated bound follows. For the second, what matters is not that the exponential term tends to 0 but that it is small *relative to the first term*: $L^2 \exp(-L/(24\log L))$ decays faster than every power of L , hence is $o(1/(L\sqrt{\log L}))$, so it can be absorbed into the tolerance of the first term, and since $\varepsilon > 0$ was arbitrary the second stated bound follows as well. $\square$

Remark 26. The window $\beta \geq 4$ is what (27) permits: at $\beta = 4$ the exponent $2 - \beta/2$ vanishes and the first term is $4\sqrt{15/\pi^2}/(2\sqrt{\log L}) = 2.4656\dots/\sqrt{\log L}$, still $o(1)$. The constant c_1 of Corollary 4 is thus an artefact of one choice of β , which is why Theorem 2 and not the numerical form is stated as the main result. Note also that letting β grow with N (§1.4) would require a fresh uniformity analysis of Lemmas 15, 17 and 19 and of Theorem 21, which we do not carry out.

9 Effectivity

Every input is effective: (M1)–(M3), (C) and (15) with explicit constants, and (P), (L) with effective (C_π, c_π) , (C_M, c_M) from the de la Vallée Poussin zero-free region. No Siegel-type or zero-density input is used, so Theorem 2 is effective in principle. What is not evaluated here is the threshold of validity: Corollary 4’s leading coefficients c_1 , c_2 are evaluated, while the constant implied by $\ll_A$ in Theorem 2 and the threshold beyond which any of these bounds holds are not. Two thresholds should be kept apart. The *normalisation* threshold, the point beyond which Proposition 8 holds, is determined by (C_M, c_M, C_π, c_π) through Lemma 24, where the explicit shape $\Lambda_2 = O(1) + 4C_M \cdot 11!c_M^{-12}$ already appears, and we make no attempt to optimise it. The *overall* threshold $N_0(\beta, \varepsilon)$ of Theorem 3 is larger: it merges that one with the thresholds of Theorem 9 and Theorem 21, and it depends in addition on the fixed β and on the requested ε , neither of which enters the normalisation.

Two features of the argument make such an evaluation unrewarding at present, and we record them because they bound what a computation could show.

1. *The two-term expression of Corollary 4 is itself larger than 1 over a very long range.* Evaluating that expression — the case $\beta = 6$, which is the only one with numerical coefficients; Theorem 2 has an unevaluated Vinogradov constant and nothing to substitute into — as if every $o(1)$ and ε were 0, the sum $c_1/(L\sqrt{\log L}) + c_2L^2 \exp(-L/(24\log L))$ first falls below 1 at $L = 3766.52\dots$, i.e. $\log_2 N = 10867.87\dots$. This is arithmetic on a formal expression and not a threshold for the theorem. At that L one has $Z_0 = \exp(L/(\log L)^3) \approx 852$ against $Y = L^6 \approx 2.9 \cdot 10^{21}$, so $u = \log Z_0/\log Y \approx 0.14 < 1$, whereas Lemmas 15, 17 and 19 all require $u \rightarrow \infty$. The quantity u at $z = Z_0$, namely $L/(6(\log L)^4)$, first reaches 1 near $L = 1.08 \cdot 10^5$, i.e. $\log_2 N \approx 3.1 \cdot 10^5$ — a factor about 29 in L beyond the point where the formal expression dips below 1.
2. *At every size at which we have evaluated it, the construction is empty.* The truncation in (10) keeps a class only when $a_Y(\ell) \leq \sqrt{Q}$, and for $\beta = 6$ the condition $Y < Q$ of (14) is $6\log L < L$, i.e. $L > 16.9988\dots$ and $\log_2 N \geq 49.05$, while $Y \leq \sqrt{Q}$ needs $\log_2 N \geq 132.51$, both figures being the continuous crossing rounded *upward*, as a sufficient threshold requires. At $N = 2^{28}$, the largest size at which §11 certifies both endpoints, $Y = L^6 \approx 8.4 \cdot 10^5 > Q$: every band element is Y -smooth, so every small-prime part exceeds $\sqrt{Q}$ and x vanishes identically. This is a statement about the sizes we evaluated,

not about all computable N . Indeed the construction is already nonempty a little past $\log_2 N = 49.05$, and a witness can be exhibited without building a band: at $N = 2^{52}$ one has $Q = 2^{26} = 67108864$, $L = 18.0218\dots$, $Y = L^6 = 34260433.5\dots < Q$, $q_1 = 67108879$ and $\text{thr} = 67108849$, and the primes 34260449 and 34260493 both lie in $(Y, Q]$, so that $2 \cdot 34260449 \in \mathcal{E}$ and $2 \cdot 34260449 \cdot 34260493 \in \mathcal{O}$, each with small-prime part $2 \leq \sqrt{Q}$; hence $B_2, R_2 > 0$ and (10) carries positive weight there. What does hold at every size we can evaluate is that $Y = L^6$ is unusable, so any evaluation of (10) at such N takes some Y far below L^6 , where the cofactor has only a handful of prime slots.

Neither feature is intrinsic. In Lemma 13 the step $|e_d| \leq \max(t_d, (Ax)_d)$ discards the cancellation between t_d and $(Ax)_d$ on the rows with a large prime factor; at the parameters we have examined the resulting bound exceeds the true ℓ^2 norm of $e|_{P+(d)>Y}$ by a large factor, which we have not measured systematically and do not quantify here. Exploiting that cancellation is the natural route to a better constant and a smaller threshold, and we have not attempted it.

10 Formalization

A Lean 4 development accompanies this paper (file `KappaZero.lean`, toolchain `leanprover/lean4:v4.33.1`, `mathlib v4.33.1`; see the data availability statement). It formalizes, with no `sorry` and with no axiom beyond `mathlib's propext`, `Classical.choice` and `Quot.sound`, a selection of the steps of the proof that are finite and exact — a selection, not all of them: part (i) of Lemma 7 and the upper bound of its part (ii), namely that an $n \leq N < (Q+1)^2$ has at most one prime factor above Q , that this prime divides n only once, and that $\lfloor N/q_1 \rfloor \leq Q$ — the Bertrand lower bound $Q/2 - 1 \leq \text{thr}$ and the logarithmic bounds of that lemma are not in the file; the exactness $d \mid n \Leftrightarrow d \mid a_Y(n)$ on Y -smooth rows together with the row identity (18) of Lemma 10, dead classes included; the large-prime removal in the $M-\pi$ form it takes at $d = 1$, that is Lemma 22; the Bonferroni identity used in Lemma 15 together with the exact size of its truncation error; the parity split and the ratio bound (26); and the containment $Ax \in K_Q$ for $x \geq 0$.

Several of these are *generic conditional* lemmas rather than statements about the band of this paper. The row identity is proved for an arbitrary finite set of squarefree integers; the parity split and its ratio bound take the two class-count identities as hypotheses, for a supplied inventory of classes, and derive the algebra from them; the cone lemma assumes a nonnegative coefficient vector. That is the stronger form to prove, and it is deliberate — but it also means the development contains no formal definition of $\kappa(N)$ for the band (2) and no theorem about it: *instantiating these lemmas at the paper's actual arithmetic model — constructing that band inside Lean, discharging the count identities for it, and assembling the instances — is not part of the formal development.*

The analytic inputs and estimates are not formalized and are not claimed to be: Mertens' three theorems, the effective prime number theorem (P), the effective Möbius bound (L), Rankin's trick, the sieve, friable and rough-Möbius estimates of Lemmas 14, 15, 16, 17 and 19, the normalisation Proposition 8, and therefore Theorems 9, 20, 21 and 2 themselves. The development certifies the combinatorial skeleton of the proof, not the proof.

11 A computational remark

Because K_Q is finitely generated, $\text{dist}_2(c_{\mathcal{O}}, K_Q)$ is the value of a nonnegative least-squares problem whose data are integral, and its square is rational. Let π be the projection of $c_{\mathcal{O}}$ on K_Q , which exists and is unique because K_Q is a closed convex cone. If $\pi = 0$ then $\text{dist}_2(c_{\mathcal{O}}, K_Q)^2 = \|c_{\mathcal{O}}\|_2^2$ is an integer. Otherwise write $\pi = \sum_{\ell} y_{\ell} A_{\ell}$ with $y \geq 0$, which is possible because $\pi \in K_Q$, and among all such representations take one whose support $S = \{\ell : y_{\ell} > 0\}$ has

fewest elements; there are finitely many generators, so such a representation exists, and $S \neq \emptyset$. Its columns are then linearly independent. For suppose $\sum_{\ell \in S} v_\ell A_\ell = 0$ with $v \neq 0$ supported on S ; replacing v by $-v$ if necessary, some $v_\ell > 0$. Then $y - tv$ represents π for every t , and it stays nonnegative for $0 \leq t \leq t^* := \min\{y_\ell/v_\ell : v_\ell > 0\} > 0$, since the coordinates with $v_\ell \leq 0$ only increase. At $t = t^*$ a further coordinate vanishes, giving a nonnegative representation of π with strictly smaller support and contradicting the choice of S . Let B collect those independent columns and y_S the corresponding coefficients. Because $y_\ell > 0$ on S , the optimality conditions for the projection give $B^\top(c_\mathcal{O} - \pi) = 0$, that is the normal equations $B^\top B y_S = B^\top c_\mathcal{O}$ with $B^\top B$ invertible, so $y_S = (B^\top B)^{-1} B^\top c_\mathcal{O}$ and $\pi = B y_S$ are rational because B and $c_\mathcal{O}$ are. Hence $\text{dist}_2(c_\mathcal{O}, K_Q)^2 = \|c_\mathcal{O} - \pi\|_2^2$ is rational, and so is $\kappa(N)^2$ for every N for which $\kappa(N)$ is defined, and both endpoints of an interval for $\kappa(N)$ can in principle be certified in exact arithmetic. That has been carried out: both endpoints are certified at every size reached, $N = 2^{16}, 2^{18}, \dots, 2^{28}$, and at $N = 2^{30}$ the lower endpoint is. This section states the inequality the certificates verify and the intervals; the verification itself is Appendix A.

A pair of certificates suffices. Write $P = c_\mathcal{E} - c_\mathcal{O}$, so that $\kappa(N) = \text{dist}_2(c_\mathcal{O}, K_Q)/\|P\|_2$ by (7). Let $q \geq 0$ be an integral vector and $D \geq 1$ an integer (a rational primal q/D), and let λ be an integral dual with $\theta_\lambda(\ell) \geq 0$ for every $\ell \in \mathcal{E}$ and $\langle c_\mathcal{O}, \lambda \rangle < 0$. Then

$$\frac{\langle c_\mathcal{O}, \lambda \rangle^2}{\|P\|_2^2 \|\lambda\|_2^2} \leq \kappa(N)^2 \leq \frac{\|D c_\mathcal{O} - Aq\|_2^2}{D^2 \|P\|_2^2}, \quad (29)$$

and both bounds are rational and checkable in integer arithmetic from the two stored vectors. The left inequality is Proposition 1 applied to $\lambda/\|\lambda\|_2$; the right is $\text{dist}_2(c_\mathcal{O}, K_Q) \leq \|c_\mathcal{O} - Aq/D\|_2$. The dual feasibility has to be verified on all of $\mathcal{E}$, not only on the columns a solver happened to touch; that check is a single pass over the band, and it is the pass that was run at every size below.

How the two vectors are obtained. A solver returns a floating-point pair, and neither half of it is a certificate until it has been rounded to integers and re-checked. The primal is taken to be Dy rounded componentwise to the nearest integer and then clamped at 0, for a per-cell integer D ; the clamp keeps $q \geq 0$, and the result leaves the right-hand side of (29) an exact rational. At the five smaller two-sided sizes the dual is built from the residual $r = c_\mathcal{O} - Ay$ of the solver's primal, for which $-r$ is dual feasible at the optimum: λ is $-r$ scaled by $K/\max_d |r_d|$, for a per-cell integer K , and rounded componentwise to the nearest integer, so that $\max_d |\lambda_d| \leq K$. At the two largest no residual is formed. There the column generation stores a floating dual w with $\langle c_\mathcal{O}, w \rangle > 0$, already repaired row-wise before it was saved — each column whose divisor sum exceeds a threshold is lowered through its divisor row of least t_d , never through the row $d = 1$, by the largest amount the columns assigned to that row require, and the threshold kept is the one giving the best floating bound — and λ is that $-w$, scaled and rounded in the same way. The stored primal is taken from the round achieving the best upper bound and the stored dual from the round achieving the best lower bound, so the two halves need not come from the same round, let alone the same vector. The lower-endpoint certificate at 2^{30} is built differently again and is described in Appendix A. Rounding can make θ_λ negative on a few columns. It is repaired at the row $d = 1$ by a scalar amount ρ , which the residual vector r above does not name: since $1 \mid \ell$ for every ℓ , replacing λ_1 by $\lambda_1 + \rho$ adds ρ to $\theta_\lambda(\ell)$ at every column at once, so $\rho = \max(0, -\min_{\ell \in \mathcal{E}} \theta_\lambda(\ell))$ restores feasibility everywhere in one step. The repair costs something when it does anything at all: since $c_\mathcal{O}(1) = R$ it moves $\langle c_\mathcal{O}, \lambda \rangle$ towards 0 by ρR , and it changes $\|\lambda\|_2^2$ by $2\rho\lambda_1 + \rho^2$, a strict increase exactly when $\rho > 0$ and the pre-repair λ_1 exceeds $-\rho/2$. Both conditions matter: the second, which for the stored coordinate $\lambda_1 + \rho$ reads $\lambda_1 + \rho > \rho/2$, holds in every certificate stored here, at all eight sizes, but the first can fail — a rounded dual that is already feasible has $\rho = 0$ and a vacuous repair, as at $e = 20$, where $\min_{\ell \in \mathcal{E}} \theta_\lambda(\ell)$ is already positive. So the certified lower endpoint is at most the solver's

recorded floating optimum — a nine-decimal ledger value at the two smallest sizes and, at the next three, the floating optimum stored in the same file as the dual; those five are the sizes at which the diagnostic clause of Appendix A compares a stored float against the interval, and at the two largest it compares none — and below it where $\rho > 0$: an observation about these certificates, not a property of the repair. The repair needs no new solve, and the vector stored in the certificate is the repaired one, already feasible as stored.

The certified intervals. For the seven two-sided cells listed below, both bounds of (29) were evaluated in exact integer arithmetic; at 2^{30} only a certified lower bound is available (described below). The endpoints below are the square roots of the two rational bounds that (29) gives for κ^2 , rounded outward at twelve decimals; the last column repeats them rounded outward at seven.

e	certified $\kappa(2^e) \geq$	certified $\kappa(2^e) \leq$	outward at 7 decimals
16	0.172389351665	0.172389351666	$0.1723893 < \kappa < 0.1723894$
18	0.140860453831	0.140860453832	$0.1408604 < \kappa < 0.1408605$
20	0.105542708383	0.105542708385	$0.1055427 < \kappa < 0.1055428$
22	0.076331197680	0.076331197683	$0.0763311 < \kappa < 0.0763312$
24	0.052762952627	0.052762952635	$0.0527629 < \kappa < 0.0527630$
26	0.034622922800	0.034623479647	$0.0346229 < \kappa < 0.0346235$
28	0.024526262835	0.024538226032	$0.0245262 < \kappa < 0.0245383$

The widths are 1.0, 1.0, 2.0, 3.0, 8.0 in units of 10^{-12} at $e = 16, \dots, 24$, then at most $5.6 \cdot 10^{-7}$ at $e = 26$ and at most $1.2 \cdot 10^{-5}$ at $e = 28$ — the last two rounded upward, so the width is below 10^{-6} at 2^{26} and below 10^{-4} at 2^{28} . How the pair was produced differs with the size, and the distinction matters only for how tight the interval is, not for whether it is certified. At $e = 16, 18, 20$ the primal is a dense nonnegative least squares over the *complete* set of distinct even- ω columns — 4414, 18 685 and 76 946 of them — so no column was excluded. At $e = 22, \dots, 28$ the full problem is out of reach and column generation was used: a restricted master over a working set, priced by one sweep over the band per round. Restricting the column set can only increase the distance, so a restricted optimum, once rounded to an integral (q, D) , is still a valid upper bound. At $e = 26$ and 28 the primal and the dual come from *different* runs: a run stopped on its wall-clock budget leaves a good primal and a dual weaker than an older converged run's, and (29) lets the two halves be taken from different vectors, each verified on its own. That pairing is what the stored certificates record, and it is frozen: the numbers above are not to be refreshed from a later re-solve without being re-certified and re-stated.

The verifier. The stored certificates are verified in exact integer arithmetic by the packet's verifier: every decision the check makes is an exact integer or rational comparison — it re-derives the cell from the exponent alone, rebuilds the band from scratch, sweeps the dual feasibility condition over *every* even- ω column of the whole band, and recomputes both rational bounds of (29). Appendix A gives those checks in full, together with the overflow guards, the format and index validation, the one floating-point step and the measured cost; the packet named in the data availability statement carries the verifier, its run commands and the frozen timing record, and its README says which outputs belong to which run.

$e = 30$: the lower endpoint only. At $N = 2^{30}$ the band has $|\mathcal{E}| = 86\,592\,333$ and $|\mathcal{O}| = 89\,520\,933$, and only the left half of (29) has been evaluated. Writing

$$J = -\langle c_{\mathcal{O}}, \lambda \rangle, \quad H = \|\lambda\|_2^2, \quad D_p = \|P\|_2^2, \quad (30)$$

so that the left half of (29) reads $\kappa(N)^2 \geq J^2/(HD_p)$ when $J > 0$, the stored run records the four integers J , H , D_p and c_6 with

$$c_6^2 H D_p < J^2 \cdot 10^{12}, \quad (31)$$

which is that half cleared of denominators and certifies $\kappa(N) > c_6/10^6$. Here $J = 4401612119558212802 > 0$, $H = 2015370564893177546709684150$, $D_p = 30382600959846$ and $c_6 = 17787$, the largest six-decimal value for which (31) holds, so

$$\kappa(2^{30}) > 0.017787.$$

We stress that (31) is an arithmetic consequence of feasibility and not a substitute for it: four integers satisfying an inequality prove nothing on their own. What makes the quantities (30) the right ones is a stored integral λ together with the sweep establishing $\theta_\lambda \geq 0$ on every even- ω column of the whole band, exactly as at the smaller sizes. The packet ships that λ — the repaired integral vector itself, not the solver’s floating separator — with the four integers beside it. No integral primal was produced at 2^{30} , so no upper endpoint is claimed there; the restricted master reports $\kappa(2^{30}) \approx 0.01779$, which is a solver value and not certified.

The construction’s own bounds. The construction (10), evaluated at the same sizes with Y chosen freely rather than as L^6 — which exceeds Q there, see §9 — gives $\kappa \leq 0.6982, 0.6130, 0.6088, 0.5841$ at $e = 16, 18, 20, 22$, attained at $Y = 3, 3, 5, 5$. These are certified too, and by the cheapest route available: they come from the integer class counts R_a, B_a through (10), so each *squared* bound is an exact rational — the bounds themselves are square roots of those rationals and are in fact irrational at all four sizes — and the four decimals printed are each the exact square root rounded upward. They exceed the certified optimum by factors between four and eight, and over the four sizes tested that factor grows with N — an observation about this range, not something §9 predicts.

One feature of these sizes is worth naming before a reader notices it. Proposition 8 asserts $R \geq 0.09N$ and $N/R \rightarrow \pi^2/(3(1 - \log 2)) = 10.7213228\dots$, both for L large; at every size computed here the band sits on the other side of those two numbers. The ratio R/N increases with N over the eight sizes, from 0.0768 at 2^{16} to 0.0834 at 2^{30} — below 0.09 throughout, and below the limiting 0.0932720\dots — so that N/R decreases from 13.03 to 11.99, approaching 10.7213\dots from above and not yet near it. There is no conflict: both statements of Proposition 8 are asymptotic, and a quantity converging to a limit from above may lie on either side of a weaker constant at any finite point.

The code, the stored primal and dual vectors, and the run commands are in the repository named in the data availability statement. None of it bears on Theorem 2 (§1.4): every size reached here lies far below the range in which the hypotheses of §5 take hold, so none of these computations can support or contradict it, and the numbers are reported only because $\kappa(N)$ is a finite conic distance whose size a reader may wish to know.

A Verification of the certificates

This appendix records what the check of §11 does: the cell binding, the format and index validation, the cache handling, the overflow guards, the one floating-point step and its check, the measured cost, and the separate construction of the lower-endpoint certificate at the largest size. Nothing here is needed to read the theorems or (29); it is the account a reader reproducing the numbers needs.

The verifier. Each certificate is stored as the two integral vectors together with the integers they need, and is re-checked by `pa_certify.py --verify`, which finds and re-checks every

stored certificate; the two largest were produced by `pa_certify_big.py`, which verifies them too. What the check derives for itself, from the exponent e alone, is the cell: $N = 2^e$, $Q = \lfloor \sqrt{N} \rfloor$, q_1 , thr , and the complete list of squarefree $d \leq Q$ from a Möbius sieve. It then refuses to proceed unless the cached cell and the stored certificate agree with that derivation in all five, so a certificate cannot be verified against a cell other than the one it names. From there the band is rebuilt from scratch and $|\mathcal{E}|$, $|\mathcal{O}|$, $c_{\mathcal{E}}$, $c_{\mathcal{O}}$ compared against the cached row counts element for element; $\theta_{\lambda} \geq 0$ is swept over *every* even- ω column of the whole band; $\langle c_{\mathcal{O}}, \lambda \rangle < 0$ and $q \geq 0$ are confirmed; and both rational bounds of (29) are recomputed. Every stored index is validated before it indexes anything and independently of the cached band — rows in $[1, Q]$, strictly increasing and squarefree; columns in $(\text{thr}, N]$, distinct, squarefree, $P^+ \leq Q$ and of even ω — both integer kernels are guarded by the actual integer sums of the vectors passed to them, and the run requires the full advertised inventory of seven cells, so a missing certificate fails rather than quietly shrinking the claim. The stored arrays must also be integer arrays whose values are preserved by conversion to signed 64-bit integers; a fractional or value-changing representation is rejected rather than converted. What is *read* rather than re-derived is the stored pair of integral vectors itself, which is the object being certified, and the cached per-row counts, which are compared against the fresh rebuild rather than trusted.

K and D are chosen per cell so that the integer kernels evaluating (29) cannot overflow, and the totals they produce are checked against exact big-integer arithmetic.

The packet distinguishes a *verification* run, which reads stored certificates and re-checks them, from a *construction* run, which re-derives and rewrites them; each run announces which it is, and only the first is a packet check.

It is cheap: one run re-checks all seven certificates in 73.2 s at four threads on the authors' machine (times on other hardware differ) — 8.6 s for the five cells $e \leq 24$ together, 13.2 s at $e = 26$ and 51.4 s at $e = 28$, band rebuild included — so solving dominates the cost of these numbers by a wide margin. Re-establishing the 2^{30} lower endpoint takes a few minutes more, the band rebuild dominating there too. The machine is specified with the figures in the reproduction packet's frozen timing record, which no reproduction step overwrites. The procedure behind the figures is fixed and the record states it: they are the wall times of one run of the command, with an empty compiler cache — a first run, which is what a reader gets, and which pays a one-time kernel compilation inside the smallest cell — and the record carries beside them the totals of the other runs of that command taken on the same machine, cold and with the cache warm, so that the spread is read off the record and not from any rule for choosing among runs. That spread is not small: the machine was not idle while any of the runs were taken, carrying unrelated work from other sessions throughout, and the band rebuild at the largest cell, which sets the total, varies between runs by more than the compilation the warm runs save. These figures therefore locate the cost on that machine and are not reproducible to the digit as the arithmetic is.

Two comparisons that are not certificate conditions. The scripts record two checks beside (29), reported apart from validity and relied on by nothing. One is a rounding consistency check: it holds a stored nine-decimal float optimum against the certified interval, and at $e = 16, 18$ the float lies outside it, a nine-decimal rounding having no reason to lie inside a twelve-decimal interval. The other is an internal target: the script behind the two largest certificates asks for a width below 10^{-6} , which holds at 2^{26} and not at 2^{28} , where the width is the one printed above — larger than the target and inside the 10^{-4} claimed there. An unmet target says that the column generation stopped further from the optimum than was hoped for, not that the certificate is defective: a certified width is whatever the stored pair gives, and both endpoints are verified in exact arithmetic either way. The exit codes report certificate validity alone, so neither comparison can be mistaken for a certificate that failed to verify.

The lower certificate at 2^{30} . Its λ is *not* built by the normalisation used at the seven two-sided cells, and the difference matters for anyone comparing the stored vector with the stored separator. Here the solver's w , which the separator holds with $\langle c_O, w \rangle > 0$, is rounded componentwise at the *fixed* scale of (31): $\lambda = \lfloor -10^{12}w \rfloor$, the sign being what makes $\langle c_O, \lambda \rangle < 0$ and hence $J > 0$. The coordinate at $d = 1$ is then *discarded*, set to zero, and rebuilt by the repair, so that λ_1 is exactly the ρ of the repair described in §11 and nothing else. Two consequences are worth stating. The scale is not chosen to fill the integer range: 10^{12} is the scale at which the four integers of (31) were computed, so a certificate built this way reproduces the J and H printed above digit for digit instead of certifying the same bound with different valid integers — and $\sum_d |\lambda_d|$ is checked against the accumulator bound at that scale rather than assumed to fit it. And because $\max_d |\lambda_d|$ is here governed by w and the scale rather than set to K , it is not the quantity the seven smaller certificates bound; a reader should expect the stored separator's $\max_d |w_d|$ and the certified vector's $\max_d |\lambda_d|$ to differ by that factor of 10^{12} , up to the componentwise rounding and the rebuilt coordinate at $d = 1$.

In this constructor — not in the check, which guards before every kernel it runs — the sweep that follows the repair is not preceded by a fresh guard, and does not need one. Write A and B for the sums of the positive and of the absolute negative coordinates of λ , so that the quantity guarded before the first sweep is $\sum_d |\lambda_d| = A + B$. The first coordinate is zero at that point, and the repair amount ρ is the negated minimum of θ_λ over the even- ω columns, each value of which is a signed subset sum of the coordinates and so at least $-B$; hence $0 \leq \rho \leq B$. After the repair every partial sum the kernel forms is ρ plus a signed subset sum of the remaining coordinates, so it lies between $\rho - B$ and $\rho + A$, hence between $-B$ and $A + B$: in magnitude it is bounded by the very $A + B$ that was guarded. The vector is in addition required to satisfy the guard again after the repair, so the certificate that leaves the constructor meets the blanket condition too.

The check re-establishes the inequality from the stored vector alone: deriving the cell, rebuilding the band, sweeping every even- ω column, recomputing J , H and D_p from it and confirming that c_6 is maximal. Its stored arrays are held to the same format requirement as the others, and no solver is involved: every decision the check makes is an exact integer or rational comparison, with the integer kernels guarded as above by the actual integer sums passed to them. Floating-point arithmetic does occur in the surrounding machinery — loading the cached arrays and the printed diagnostics, where no verdict depends on it, and one square root setting a loop bound in the sieve, whose truncation the verifier checks to coincide with the exact integer square root on the whole range where the sieve is used.

References

- [1] K. Alladi, *Asymptotic estimates of sums involving the Moebius function*, J. Number Theory **14** (1982), 86–98.
- [2] K. Alladi, *Asymptotic estimates of sums involving the Moebius function. II*, Trans. Amer. Math. Soc. **272** (1982), 87–105.
- [3] A. R. Booker and T. D. Browning, *Square-free values of reducible polynomials*, Discrete Anal. **2016**, Paper No. 8.
- [4] Z. E. Brady, *Sieves and iteration rules*, Ph.D. thesis, Stanford University, 2017.
- [5] Z. E. Brady, *A semidefinite framework for the sieve*, preprint (2021), [arXiv:2112.02722](https://arxiv.org/abs/2112.02722).
- [6] R. de la Bretèche and G. Tenenbaum, *Friable averages of oscillating arithmetic functions*, in: Number Theory in Memory of Eduard Wirsing (H. Maier, J. Steuding and R. Steuding,

- eds.), Springer, Cham, 2023, pp. 43–71; preprint [arXiv:2207.04777](https://arxiv.org/abs/2207.04777), whose listing carries the title *Friable averages of oscillating multiplicative functions*.
- [7] K. Ford and J. Maynard, *On the theory of prime producing sieves*, preprint (2024), [arXiv:2407.14368](https://arxiv.org/abs/2407.14368).
 - [8] C. S. Franze, *Sifting limits for the $\Lambda^2\Lambda^-$ sieve*, J. Number Theory **131** (2011), 1962–1982.
 - [9] J. B. Friedlander, *Integers free from large and small primes*, Proc. London Math. Soc. (3) **33** (1976), 565–576.
 - [10] J. Friedlander and H. Iwaniec, *Opera de Cribro*, Amer. Math. Soc. Colloq. Publ. **57**, Amer. Math. Soc., Providence, RI, 2010.
 - [11] D. R. Heath-Brown, *Lectures on sieves*, preprint (2002), [arXiv:math/0209360](https://arxiv.org/abs/math/0209360).
 - [12] A. Hildebrand and G. Tenenbaum, *Integers without large prime factors*, J. Théor. Nombres Bordeaux **5** (1993), 411–484.
 - [13] D. H. J. Polymath, *Variants of the Selberg sieve, and bounded intervals containing many primes*, Res. Math. Sci. **1** (2014), Art. 12.
 - [14] J. B. Rosser and L. Schoenfeld, *Approximate formulas for some functions of prime numbers*, Illinois J. Math. **6** (1962), 64–94.
 - [15] É. Saias, *Entiers sans grand ni petit facteur premier. I*, Acta Arith. **61** (1992), 347–374.
 - [16] É. Saias, *Entiers sans grand ni petit facteur premier. II*, Acta Arith. **63** (1993), 287–312.
 - [17] V. Soltan, *Moreau-type characterizations of polar cones*, Linear Algebra Appl. **567** (2019), 45–62; preprint [arXiv:1810.04206](https://arxiv.org/abs/1810.04206), 2018, whose numbering the theorem cited above follows.
 - [18] G. Tenenbaum, *Cribler les entiers sans grand facteur premier*, Philos. Trans. Roy. Soc. London Ser. A **345** (1993), 377–384.
 - [19] G. Tenenbaum, *Introduction to Analytic and Probabilistic Number Theory*, 3rd ed., Grad. Stud. Math. **163**, Amer. Math. Soc., Providence, RI, 2015.
 - [20] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd ed., revised by D. R. Heath-Brown, Oxford Univ. Press, Oxford, 1986.
 - [21] T. S. Trudgian, *Updating the error term in the prime number theorem*, Ramanujan J. **39** (2016), 225–234.

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Data availability statement. The audit code, stored outputs, and Lean 4 development accompanying this manuscript are openly available at <https://github.com/soke91/arithmetic-convolution-fields>, in the directory P7-parity-separation-tends-to-zero/, at the fixed release tag P7-v1. No data derived from human or animal subjects were used.